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CCR-Motivated Infrared Cutoff and CMB Power Spectrum Suppression

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Abstract. We test the Conditional Cosmological Recurrence (CCR) framework against Planck 2018 CMB data by identifying an infrared cutoff k_c in the primordial power spectrum from the finite Hilbert-space dimension D of the de Sitter causal patch during inflation. The cutoff is not a free parameter: it is a deterministic function of D alone, fixed by the Gibbons–Hawking entropy bound and standard thermal-history inputs. For $\ln D \sim 10^{14}$, the predicted scale $k_c \sim 10^{-4}$ Mpc^{-1} falls in the window responsible for the observed low-ℓ CMB power deficit. We implement the CCR-modified spectrum in the CAMB Boltzmann code and constrain the model parameters $\log_{10}(\ln D)$ and shape parameter α using Markov Chain Monte Carlo sampling with Planck 2018 TT+TE+EE likelihoods. The data do not exclude the CCR framework but do not prefer it over ΛCDM: the Savage–Dickey Bayes factor is $\ln B_{01} = 0.06$, the BIC penalises the two extra parameters ($\Delta\mathrm{BIC} = +13$), and the posterior on $\ln D$ is approximately uniform over the prior range. This null result is itself informative: it establishes that a 3σ detection requires $\ln D \lesssim 10^{13}$ (equivalently $k_c \gtrsim 7 \times 10^{-4}$ Mpc^{-1}), a threshold accessible to the combination of LiteBIRD polarisation and Planck temperature data. The CCR framework simultaneously predicts the tensor-to-scalar ratio $r = 16/(A_s \ln D)$, providing a two-observable consistency test with future B-mode experiments.

Keywords: CMB power spectrum; primordial power spectrum; inflation; infrared cutoff; Bayesian inference; holographic entropy

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1 Introduction

The angular power spectrum C_ℓ of the cosmic microwave background (CMB) provides one of the most precise tests of the standard ΛCDM cosmological model. At multipoles $\ell \gtrsim 30$, the agreement between the Planck 2018 observations [13] and the six-parameter ΛCDM prediction is remarkable. At large angular scales ($\ell \lesssim 30$), however, the observed power falls systematically below the theoretical expectation—a deficit that has persisted across COBE, WMAP, and Planck data releases and is robust across independent component-separation methods. The reconstructed primordial power spectrum shows a $\sim 2\sigma$ suppression at $k \lesssim 10^{-3}\,\mathrm{Mpc}^{-1}$ [12], and the phenomenological infrared cutoff model of Contaldi et al. [2] yields a $\Delta\chi^2 \simeq 5.3$ improvement over ΛCDM with a best-fit cutoff scale $k_c \simeq 4.9 \times 10^{-4}\,\mathrm{Mpc}^{-1}$.

Several physical mechanisms have been proposed to account for this low-ℓ deficit. Contaldi et al. [2] introduced a kinetic-energy dominated phase preceding slow-roll inflation, which suppresses long-wavelength fluctuations; however, the cutoff scale k_c is a free parameter with no connection to a fundamental theory. Loop quantum cosmology replaces the initial singularity with a quantum bounce that modifies the primordial spectrum at large scales [15, 16], but involves a different physical mechanism and makes no direct contact with information-theoretic bounds. The deficit may also be a statistical fluctuation due to cosmic variance, which fundamentally limits the number of independent modes at low ℓ: the probability of observing the measured quadrupole–octupole deficit under ΛCDM is approximately 5–10% [13, 17]. This is not a physical explanation but rather the null hypothesis against which any model must be tested. A common feature of all existing proposals is that the suppression scale k_c—when present—is either a phenomenological free parameter or arises from model-specific initial conditions, without derivation from an independently published, falsifiable theoretical framework.

In this paper we show that a natural infrared cutoff in the primordial power spectrum emerges from the Conditional Cosmological Recurrence (CCR) framework developed in ref. [1]. The CCR framework rests on a single foundational assumption: the causal patch accessible to a comoving observer during de Sitter inflation admits a *finite*-dimensional operational Hilbert space, with dimension D bounded by the holographic entropy of the patch, $\ln D \leq A/(4\,l_P^2)$, where A is the area of the de Sitter event horizon and l_P is the Planck length. This bound is motivated by the Bekenstein–Hawking entropy [4, 5] and the Gibbons–Hawking entropy of de Sitter space [6]. The resulting dimension D is astronomically large ($D \sim \exp(10^{14})$ for inflationary parameters) but strictly finite.

The central result of the CCR framework is that finite D implies unitary quantum dynamics on a finite-dimensional Hilbert space, which in turn guarantees almost-periodic evolution and conditional cosmological recurrences—provided the assumption of finite D holds. Crucially, the framework is explicitly falsifiable: if a future theory of quantum gravity establishes that the Hilbert space of a causal patch is infinite-dimensional, the CCR framework and all its predictions are invalidated.

For the purposes of this paper, the relevant consequence of finite D is not the recurrence phenomenon itself, but the information-theoretic restriction it imposes on the quantum field theory within the causal patch. A finite-dimensional Hilbert space can support only a finite number of independent quantum field modes; modes with physical wavelength exceeding the de Sitter horizon H_{inf}^{-1} cannot be operationally resolved within the patch. This truncation of the mode expansion generates an effective infrared cutoff at $k_{c,\,\mathrm{phys}} = H_{\mathrm{inf}}$, which suppresses primordial power at large scales precisely where the observed deficit occurs.

Unlike phenomenological models, the cutoff k_c in our approach is not a free parameter: it is a deterministic function of the single quantity D, given standard thermal-history inputs (N, $T_{\rm reh}$, g_{*S}). The modified primordial power spectrum takes the form $\mathcal{P}_{\rm CCR}(k) = \mathcal{P}_{\Lambda\rm CDM}(k)\,[1 - \exp(-(k/k_c)^\alpha)]$, where α controls the sharpness of the transition.

This paper represents the first attempt to confront the CCR framework with observational data, by deriving constraints on D from the Planck 2018 CMB power spectrum.

Scope and contribution. This paper represents the first attempt to confront the CCR framework with observational data. We emphasise at the outset that the CCR framework predicts the *scale* of the cutoff k_c but not the *shape* of the transition, which is controlled by the parameter α. Deriving α from first principles requires a microphysical calculation of mode generation at horizon crossing in a finite-dimensional Hilbert space—a task that lies beyond the scope of this phenomenological study and is deferred to future work. We therefore treat α as a free parameter, so that the CCR model has one additional degree of freedom relative to ΛCDM (α) rather than two (k_c and α), since k_c is derived from $\ln D$.

Beyond constraining $\ln D$, this work establishes the complete observational infrastructure—modified Boltzmann code, MCMC pipeline, and forecast methodology—needed to test the CCR framework against current and future CMB data. The null result obtained here is itself informative: it quantifiesthe detection threshold ($\ln D \lesssim 10^{13}$ for a 3σ detection) that future experiments must surpass, setting the stage for LiteBIRD and CMB-S4. We emphasise that this work represents a *minimal phenomenological implementation* of the CCR framework; a first-principles derivation of the transition shape α from the Mukhanov–Sasaki equation in a finite-dimensional Hilbert space is the natural next step.

The paper is organised as follows. Section 2 derives the infrared cutoff $k_c = k_c(D)$ from the CCR holographic bound and analyses its sensitivity to thermal-history parameters. Section 3 describes the computational pipeline: the implementation of $\mathcal{P}_{\rm CCR}(k)$ in the CAMB Boltzmann code [14], the Planck 2018 likelihood, and the MCMC sampling strategy. Section 4 presents posterior constraints on D and α, the Bayes factor relative to ΛCDM, and a comparison with the Contaldi et al. best-fit. Section 5 discusses the physical implications, limitations, and prospects for future tests.

2 Theoretical Framework: From Finite Hilbert Space to an IR Cutoff

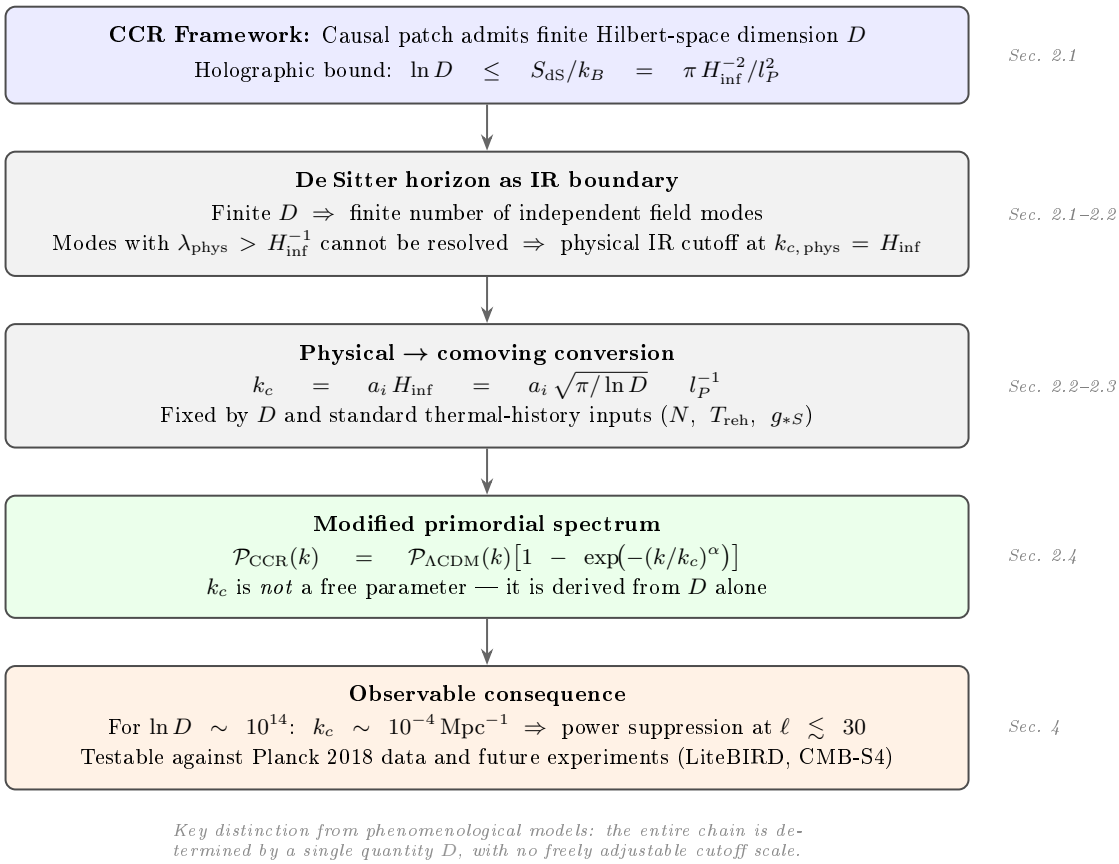

Figure 1. Logical chain from the CCR framework to the observable CMB power suppression. The infrared cutoff k_c is not a free parameter but is uniquely determined by the Hilbert-space dimension D through the Gibbons–Hawking entropy bound.

In this section we obtain the infrared cutoff k_c from the CCR framework [1]. The argument proceeds in three steps that each rest on well-established physics: (i) during inflation the de Sitter event horizon defines the causal boundary of the patch, and the finite information capacity of this patch—encoded in the Hilbert-space dimension D—truncates the support of the quantum mode expansion at a physical scale set by $H_{\rm inf}$; (ii) the CCR holographic bound uniquely relates $H_{\rm inf}$ to D; (iii) converting to comoving wavenumbers via the inflationary expansion yields the observable cutoff $k_c = k_c(D)$. Unlike previous IR-cutoff models [2], k_c is not a free parameter but a derived quantity, uniquely fixed by D.

2.1 The de Sitter Horizon as a Natural IR Cutoff

During a quasi-de Sitter inflationary epoch with Hubble parameter $H_{\rm inf}$, the causal (event) horizon of a comoving observer is located at physical distance

$$R_{\rm dS} = H_{\rm inf}^{-1} \,. \tag{2.1}$$

In quantum field theory on de Sitter space, the natural vacuum state is the Bunch–Davies vacuum [3], in which sub-horizon modes ($k_{\rm phys} \gg H_{\rm inf}$) are well-described by Minkowski-space plane waves, while super-horizon modes ($k_{\rm phys} \ll H_{\rm inf}$) freeze out after horizon crossing and are subsequently stretched to cosmological scales [7–10]. In the standard treatment with an infinite-dimensional Hilbert space, the mode expansion extends to arbitrarily long wavelengths, and super-horizon modes are set by initial conditions.

In the CCR framework, however, the Hilbert space of the causal patch has *finite* dimension D. The consequences of this finiteness for the quantum field theory within the patch can be stated precisely in effective-field-theory (EFT) language. A scalar field φ in the Bunch–Davies vacuum, restricted to a spatial volume $V \sim H_{\rm inf}^{-3}$, admits a discrete set of mode functions $\{u_{\mathbf{k}}\}$ satisfying the Klein–Gordon equation with appropriate boundary conditions. In the standard treatment the mode sum extends over an infinite tower of wavenumbers, each contributing an independent pair of creation and annihilation operators $(\hat{a}_{\mathbf{k}},\, \hat{a}^\dagger_{\mathbf{k}})$ obeying the canonical commutation relation $[\hat{a}_{\mathbf{k}},\, \hat{a}^\dagger_{\mathbf{k}'}] = \delta_{\mathbf{k}\mathbf{k}'}$. Each such pair spans a bosonic Fock space of unbounded dimension; accommodating $\mathcal{N}$ independent modes therefore requires a Hilbert space of dimension at least $d_{\rm min} \sim \mathcal{N}^p$, where $p \geq 1$ depends on the occupancy truncation [29].

The CCR constraint $\dim \mathcal{H} = D < \infty$ imposes an upper bound on the number of independent mode pairs that can be faithfully represented: the system admits at most $\mathcal{N}_{\rm max} \sim (\ln D)^{1/p}$ independent degrees of freedom. Modes with physical wavelength $\lambda_{\rm phys} > H_{\rm inf}^{-1}$ would require additional degrees of freedom beyond this budget; they are not dynamically excluded, but rather *have no independent state-space support* within the finite Hilbert space of the patch. This is an *information-theoretic* truncation, analogous to the UV cutoff that a lattice spacing imposes on a field theory—here applied at the infrared end by the finite entropy capacity of the de Sitter horizon.

The horizon scale $H_{\rm inf}^{-1}$ therefore plays a dual role: it is both the classical causal boundary and the IR resolution limit of the finite-dimensional quantum theory. We emphasise that this identification does not modify General Relativity or the inflationary dynamics; it restricts only the *information content* of the quantum state accessible to a single causal-patch observer. The resulting effective IR cutoff is a *controlled approximation*: it is exact in the limit that the Gibbons–Hawking entropy saturates the holographic bound (eq. (2.5)), and it reduces smoothly to the standard infinite-dimensional treatment in the limit $D \to \infty$. The horizon-crossing condition

$$k_{\rm phys} \;=\; H_{\rm inf} \tag{2.2}$$

therefore defines the physical scale below which the finite-dimensional Hilbert space can no longer support independent quantum fluctuations. This identification is the central physical input of the CCR approach and distinguishes it from both the standard infinite-dimensional treatment and from ad hoc phenomenological cutoffs.

We note that the identification of the truncation scale with exactly $H_{\rm inf}$ is a simplifying assumption. In principle, the finite-dimensional mode budget could saturate at a scale $\beta H_{\rm inf}$ with $\beta = \mathcal{O}(1)$, which would shift $k_c \to \beta\, k_c$ throughout. Since k_c enters logarithmically in the observable constraints (via $\ln D$), an $\mathcal{O}(1)$ prefactor does not qualitatively alter our results, but it does introduce a systematic uncertainty that should be addressed in a first-principles derivation of the mode truncation.

2.2 Connecting $H_{\rm inf}$ to the Hilbert-Space Dimension D

The CCR framework [1] starts from assumption (A1): a causal patch of area A obeys the Bekenstein–Hawking entropy bound [4, 5],

$$S_{\rm max} \;\leq\; \frac{k_B\, A}{4\, l_P^2}\,, \tag{2.3}$$

which implies a finite Hilbert-space dimension:

$$\ln D \;\leq\; \frac{S_{\rm max}}{k_B}\,. \tag{2.4}$$

For the de Sitter patch with horizon radius $R_{\rm dS} = H_{\rm inf}^{-1}$, the horizon area is $A = 4\pi R_{\rm dS}^2 = 4\pi H_{\rm inf}^{-2}$, and the Gibbons–Hawking entropy [6] saturates the bound:

$$S_{\rm dS} \;=\; \frac{k_B\, \pi}{H_{\rm inf}^2\, l_P^2} \;=\; \ln D\,. \tag{2.5}$$

Solving for $H_{\rm inf}$:

$$H_{\rm inf} \;=\; \frac{1}{l_P}\sqrt{\frac{\pi}{\ln D}}\,. \tag{2.6}$$

This is the key result: the Hubble parameter during inflation is uniquely determined by the Hilbert-space dimension D of the causal patch. Combined with eq. (2.2), the physical IR cutoff is

$$k_{c,\,\rm phys} \;=\; H_{\rm inf} \;=\; \frac{1}{l_P}\sqrt{\frac{\pi}{\ln D}}\,. \tag{2.7}$$

Physical interpretation. Equation (2.6) states that a patch with a larger Hilbert-space dimension D (higher entropy capacity) corresponds to a *lower* inflationary Hubble rate, and hence a *larger* causal horizon $R_{\rm dS} = H_{\rm inf}^{-1}$. A larger horizon admits longer-wavelength modes within the finite-dimensional Hilbert space, so the IR cutoff $k_{c,\,\rm phys}$ shifts to smaller values as D increases. In the limit $D \to \infty$ ($\ln D \to \infty$), $H_{\rm inf} \to 0$ and the cutoff disappears: the Hilbert space becomes infinite-dimensional, the information-theoretic truncation is removed, and the standard ΛCDM power spectrum is recovered.

Saturation assumption. Equation (2.5) uses equality, $\ln D = S_{\rm dS}/k_B$, corresponding to the assumption that the de Sitter entropy saturates the holographic bound. If the bound is not saturated, i.e. $\ln D = \xi\, S_{\rm dS}/k_B$ with $0 < \xi \leq 1$, then $H_{\rm inf} \to H_{\rm inf}/\sqrt{\xi}$ and correspondingly $k_c \to k_c/\sqrt{\xi}$: the cutoff shifts to *larger* values, producing *stronger* suppression. The saturated case $\xi = 1$ therefore yields the most conservative CCR prediction. A detailed discussion is given in appendix A.

2.3 From Physical to Comoving Wavenumber

Observational CMB quantities are expressed in terms of *comoving* wavenumbers, which are constant during the expansion. The comoving IR cutoff is obtained by multiplying the physical cutoff by the scale factor a_i at the onset of inflation (we normalise $a_0 = 1$ today):

$$k_c \;=\; a_i\, k_{c,\,\rm phys} \;=\; \frac{a_i}{l_P}\sqrt{\frac{\pi}{\ln D}}\,. \tag{2.8}$$

The scale factor a_i. Under the standard thermal history [10, 11], the scale factor at the onset of inflation is related to the present CMB temperature T_0, the reheating temperature $T_{\rm reh}$, and the number of inflationary e-folds N through entropy conservation. The comoving entropy density $s \propto g_{*S}\, T^3\, a^3$ is conserved after reheating, giving

$$\frac{a_{\rm reh}}{a_0} \;=\; \left(\frac{g_{*S,0}}{g_{*S,\rm reh}}\right)^{1/3} \frac{T_0}{T_{\rm reh}}\,, \tag{2.9}$$

where g_{*S} denotes the effective number of relativistic degrees of freedom contributing to entropy. Since $a_i = a_{\rm reh}\, e^{-N}$, we obtain

$$a_i \;=\; e^{-N} \left(\frac{g_{*S,0}}{g_{*S,\rm reh}}\right)^{1/3} \frac{T_0}{T_{\rm reh}}\,. \tag{2.10}$$

We adopt $g_{*S,\rm reh} = 106.75$ (the full Standard Model value at $T \gg 100\,\mathrm{GeV}$) and $g_{*S,0} = 3.938$ (photons plus three neutrino species after e^+e^- annihilation) [11], yielding a correction factor $(g_{*S,0}/g_{*S,\rm reh})^{1/3} \simeq 0.333$. We fix the fiducial values $N = 60$ and $T_{\rm reh} = 10^{15}\,\mathrm{GeV}$, consistent with standard slow-roll inflation and grand-unification-scale reheating [9].

The key equation. Substituting eq. (2.10) into eq. (2.8), the comoving IR cutoff predicted by the CCR framework is

$$\boxed{k_c \;=\; \frac{e^{-N}}{l_P} \left(\frac{g_{*S,0}}{g_{*S,\rm reh}}\right)^{1/3} \frac{T_0}{T_{\rm reh}} \sqrt{\frac{\pi}{\ln D}}} \tag{2.11}$$

This is the central result of this section. k_c is *not* a free parameter: it is a derived function of the single fundamental quantity D, given fixed thermal-history parameters. For the fiducial values above and $T_0 = 2.7255\,\mathrm{K}$:

$$k_c \;\simeq\; 1.6 \times 10^{-4}\,\mathrm{Mpc}^{-1} \qquad \text{for} \quad \ln D \;\simeq\; 2.0 \times 10^{14}\,. \tag{2.12}$$

This value lies within the observable window ($k_c \sim 10^{-4}$–$10^{-3}\,\mathrm{Mpc}^{-1}$) relevant for low-$\ell$ CMB suppression. The corresponding inflationary Hubble rate is $H_{\rm inf} \simeq 1.5 \times 10^{12}\,\mathrm{GeV}$, a factor ~ 40 below the observational upper bound $H_{\rm inf} < 6 \times 10^{13}\,\mathrm{GeV}$ set by Planck tensor-mode constraints [12].

Table 1 lists the derived CCR quantities for a range of $\ln D$ values, demonstrating the smooth dependence of k_c on the Hilbert-space dimension.

2.4 The Just-Enough-Inflation Condition

For k_c to produce an observable suppression in the Planck low-ℓ data, it must be comparable to the present Hubble scale $a_0 H_0 \simeq 2.3 \times 10^{-4}\, h\,\mathrm{Mpc}^{-1}$. From eq. (2.11), this requires $N \approx 60$ e-folds of inflation, precisely the minimum needed to solve the horizon and flatness problems [9]: the *just-enough-inflation* condition. If $N \gg 60$, the comoving cutoff is redshifted beyond the observable universe; if $N \ll 60$, it moves to $\ell > 30$, conflicting with the acoustic-peak structure.

The observability of k_c for $N \approx 60$ does not require an exponentially narrow range of e-folds; it occupies a non-negligible fraction of the physically motivated N range, as quantified below.

$\ln D$	k_c [Mpc^{-1}]	$H_{\rm inf}$ [GeV]	$R_{\rm dS}$ [m]	$R_{\rm dS}$ [l_P]
1.0×10^{13}	7.33×10^{-4}	6.84×10^{12}	2.88×10^{-29}	1.78×10^{6}
5.0×10^{13}	3.28×10^{-4}	3.06×10^{12}	6.45×10^{-29}	3.99×10^{6}
1.0×10^{14}	2.32×10^{-4}	2.16×10^{12}	9.12×10^{-29}	5.64×10^{6}
2.0×10^{14}	1.64×10^{-4}	1.53×10^{12}	1.29×10^{-28}	7.98×10^{6}
5.0×10^{14}	1.04×10^{-4}	9.68×10^{11}	2.04×10^{-28}	1.26×10^{7}
1.0×10^{15}	7.33×10^{-5}	6.84×10^{11}	2.88×10^{-28}	1.78×10^{7}

Table 1. Derived CCR quantities for fiducial parameters $N = 60$, $T_{\rm reh} = 10^{15}$ GeV, $g_{*S,\rm reh} = 106.75$, $g_{*S,0} = 3.938$. The Contaldi et al. [2] best-fit value $k_c \simeq 4.9 \times 10^{-4}\,\mathrm{Mpc}^{-1}$ corresponds to $\ln D \simeq 2.2 \times 10^{13}$ in the CCR framework. All values of $H_{\rm inf}$ lie well below the Planck tensor-mode upper bound $H_{\rm inf} < 6 \times 10^{13}$ GeV.

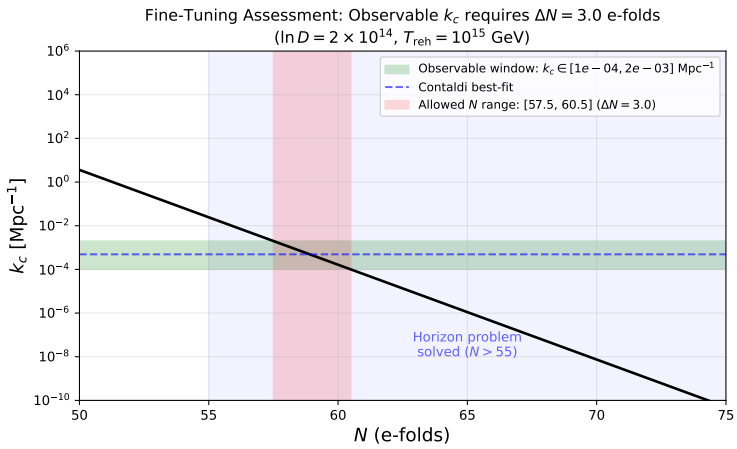

Figure 2. Observable bandwidth in N. The shaded region marks $N \in [57.5, 60.5]$ for which k_c falls in the observable window $k_c \in [10^{-4}, 2 \times 10^{-3}]\,\mathrm{Mpc}^{-1}$, at fixed $\ln D = 2 \times 10^{14}$ and $T_{\rm reh} = 10^{15}$ GeV. Horizontal dashed lines mark the observable-window boundaries. The range $\Delta N \simeq 3$ represents $\sim 5\%$ of the physically motivated $N \in [55, 70]$ prior.

Quantitative assessment. To quantify the sensitivity of k_c to N, we define the observable window as $k_c \in [10^{-4}, 2 \times 10^{-3}]\,\mathrm{Mpc}^{-1}$, corresponding to suppression at multipoles $\ell \lesssim 30$. For the fiducial values $\ln D = 2 \times 10^{14}$ and $T_{\rm reh} = 10^{15}$ GeV, this window is satisfied for $N \in [57.5, 60.5]$, a range of $\Delta N \simeq 3$ e-folds (see figure 2). Since the horizon and flatness problems independently require $N \gtrsim 55$, the observable range of k_c occupies a finite fraction ($\Delta N/N \sim 5\%$) of the physically motivated N range, rather than demanding a precise value.

Conversely, for $\sim 95\%$ of the physically motivated N range the cutoff is redshifted beyond the observable window, so the default expectation is that the CCR suppression is undetectable. Observability therefore requires a moderate coincidence between the duration of inflation and the horizon scale — a point we regard as a quantitative fine-tuning caveat rather than a fundamental obstacle, but one that should be borne in mind when interpreting any future detection.

Degeneracy and sensitivity to thermal-history parameters. Equation (2.11) shows that k_c depends on N, $T_{\rm reh}$, and g_{*S} in addition to D, introducing degeneracies in the (D, N, $T_{\rm reh}$) parameter space. In this work we break these degeneracies by fixing $N = 60$ and

Table 2. Sensitivity of k_c [Mpc^{-1}] to the number of e-folds N and reheating temperature $T_{\rm reh}$, for fixed $\ln D = 2 \times 10^{14}$. The observable window $k_c \sim 10^{-4}$–10^{-3} Mpc^{-1} is highlighted.

N	10^{10} GeV	10^{12} GeV	10^{14} GeV	10^{15} GeV	10^{16} GeV
50	3.6×10^{5}	3.6×10^{3}	3.6×10^{1}	3.6×10^{0}	3.6×10^{-1}
55	2.4×10^{3}	2.4×10^{1}	2.4×10^{-1}	2.4×10^{-2}	2.4×10^{-3}
60	1.6×10^{1}	1.6×10^{-1}	$\mathbf{1.6 \times 10^{-3}}$	$\mathbf{1.6 \times 10^{-4}}$	1.6×10^{-5}
65	1.1×10^{-1}	1.1×10^{-3}	1.1×10^{-5}	1.1×10^{-6}	1.1×10^{-7}
70	7.4×10^{-4}	7.4×10^{-6}	7.4×10^{-8}	7.4×10^{-9}	7.4×10^{-10}

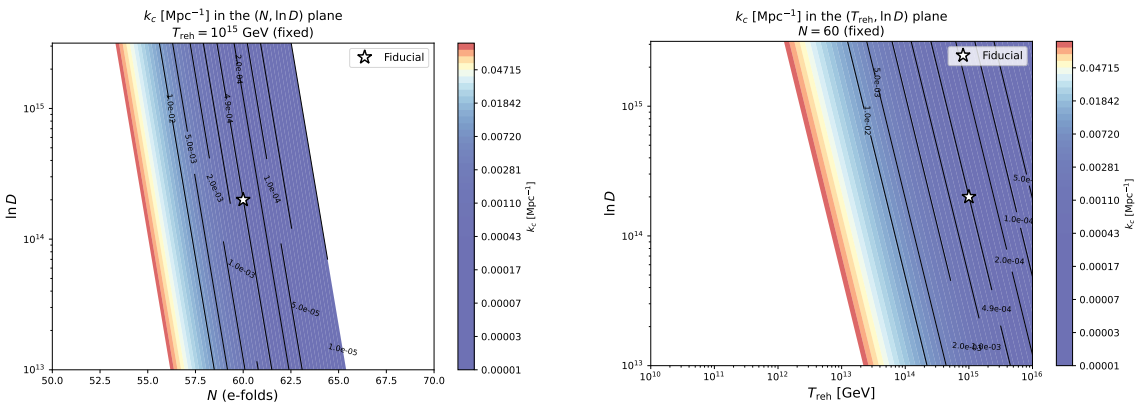

Figure 3. Contour plots of k_c [Mpc^{-1}] in the (N, ln D) plane (*left*) and the ($T_{\rm reh}$, ln D) plane (*right*). The green band marks the observable window $k_c \in [10^{-4}, 10^{-3}]$ Mpc^{-1}. The right panel is evaluated at fixed $N = 60$; the left panel at fixed $T_{\rm reh} = 10^{15}$ GeV.

$T_{\rm reh} = 10^{15}$ GeV, reducing k_c to a function of D alone. We have verified that the results are robust under variations of N and $T_{\rm reh}$ within physically motivated ranges: table 2 presents k_c for $N \in [50, 70]$ and $T_{\rm reh} \in [10^{10}, 10^{16}]$ GeV at fixed $\ln D = 2 \times 10^{14}$. The observable window $k_c \sim 10^{-4}$–10^{-3} Mpc^{-1} is achieved for a broad range of (N, $T_{\rm reh}$) combinations (see figure 3). A full marginalisation over N and $T_{\rm reh}$ with physically motivated priors is a natural extension of this analysis, but is beyond the scope of the present phenomenological study.

2.5 The CCR-Modified Primordial Power Spectrum

The finite Hilbert-space dimension D suppresses primordial fluctuations at $k < k_c$: the finite information capacity of the causal patch truncates the mode expansion, so that modes with wavelengths exceeding $H_{\rm inf}^{-1}$ have no independent degrees of freedom in which to carry quantum fluctuations. We model this suppression as:

$$\mathcal{P}_{\rm CCR}(k) \;=\; A_s \left(\frac{k}{k_*}\right)^{n_s - 1} \left[1 - \exp\!\left(-\left(\frac{k}{k_c}\right)^{\alpha}\right)\right], \tag{2.13}$$

where A_s is the scalar amplitude, $k_* = 0.05\,\mathrm{Mpc}^{-1}$ is the pivot scale, n_s is the scalar spectral index, and $\alpha > 0$ controls the sharpness of the transition.

Limiting behaviour. For $k \gg k_c$ the exponential vanishes and the standard Harrison–Zel'dovich spectrum is recovered. For $D \to \infty$ (equivalently $k_c \to 0$), the full ΛCDM power

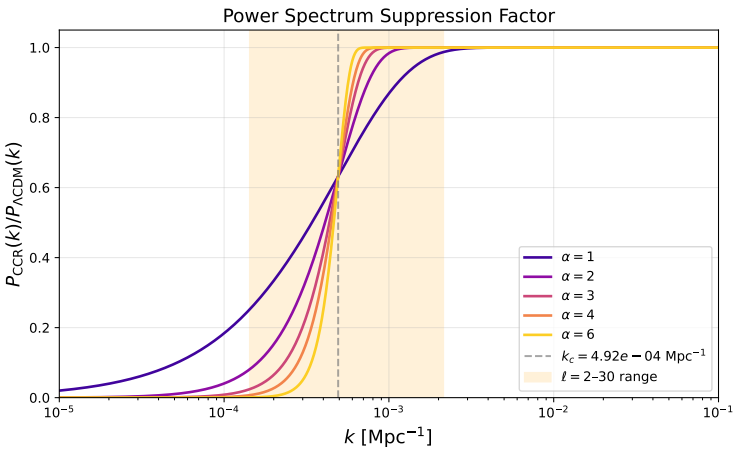

Figure 4. Suppression factor $\mathcal{P}_{\rm CCR}(k)/\mathcal{P}_{\Lambda\rm CDM}(k)$ as a function of k/k_c for shape parameters $\alpha = 1, 2, 3, 4, 6$ (see eq. (2.13)). Larger α produces a sharper transition; $\alpha = 1$ gives a gradual exponential suppression. All curves are evaluated at the fiducial cutoff $k_c = 1.6 \times 10^{-4}\,\mathrm{Mpc}^{-1}$.

spectrum is recovered at all scales, confirming that the CCR-modified spectrum is a one-parameter deformation of the standard model with a well-defined limit. We have verified this recovery numerically (see section 4).

The shape parameter α. The CCR framework predicts the existence of the cutoff scale k_c but does not uniquely fix the shape of the transition in this phenomenological treatment. The shape depends on the detailed microphysics of mode generation at horizon crossing. At the level of phenomenological template behaviour, different values of α produce qualitatively distinct suppression profiles: $\alpha = 1$ gives a gradual exponential suppression; $\alpha = 2$ produces a Gaussian-like cutoff; larger values $\alpha \gtrsim 4$ approach a sharp step function.

We treat α as a free parameter with flat prior $\alpha \in [1, 6]$. The lower bound ensures a smooth, physically motivated transition, while the upper bound avoids numerical artefacts in the CAMB Boltzmann integration (see section 3). Figure 4 shows the suppression factor $\mathcal{P}_{\rm CCR}/\mathcal{P}_{\Lambda\rm CDM}$ for several values of α: the curves differ appreciably within the $\ell = 2$–30 wavenumber range, indicating that CMB data can discriminate between different transition shapes.

Remark on the parameter count. It is important to distinguish between the number of *sampled* parameters and the number of *free* parameters. The CCR model samples two parameters beyond ΛCDM: $\log_{10}(\ln D)$ and α. However, k_c is not independently free—it is a deterministic function of $\ln D$ via eq. (2.8). In contrast, the Contaldi et al. model treats both k_c and α as independent free parameters. The CCR framework therefore replaces one phenomenological degree of freedom (k_c) with a theoretically motivated one ($\ln D$), while simultaneously generating a prediction for the tensor-to-scalar ratio (section 2.7) that is absent from the purely phenomenological approach. The shape parameter α remains free because it encodes the microphysics of the cutoff transition, which the CCR framework does not determine at this level. A first-principles derivation of α from the Mukhanov–Sasaki equation in a finite-dimensional Hilbert space is the natural next step, but lies beyond the scope of this work.

	Contaldi et al. (2003)	This work (CCR)
IR cutoff origin	Kinetic domination (ad hoc)	De Sitter horizon $H_{\rm inf}$
$H_{\rm inf}$ determined by	Not addressed	$\ln D$ via Gibbons–Hawking
Sampled parameter	k_c (free)	D (fundamental)
k_c status	Free parameter	Derived: $k_c = k_c(D)$
Independent framework	No	Yes (CCR, [1])
α treatment	Fixed (3.35)	Free, $\alpha \in [1, 6]$
Entropy correction	Not applicable	g_{*S} included
Observational constraints on	k_c	D and $H_{\rm inf}$

Table 3. Comparison between the Contaldi et al. [2] phenomenological IR cutoff and the CCR-motivated cutoff of this work. The key distinction is that k_c is a free parameter in the former and a derived quantity in the latter.

2.6 Comparison with Contaldi et al. (2003)

The functional form of eq. (2.13) is formally identical to the exponential-cutoff template of ref. [2]. The two approaches differ, however, in a fundamental respect summarised in table 3.

In ref. [2], k_c is a free phenomenological parameter introduced to improve the fit to low-ℓ data, with physical motivation limited to a model-specific kinetic-domination stage before slow-roll inflation. There is no connection to an independently published theoretical framework, and no prediction for any other observable.

In this work, k_c is entirely determined by D through eq. (2.11). The CCR framework [1] provides an independently peer-reviewed foundation: the finite information capacity of the de Sitter patch sets $H_{\rm inf}$ via the Gibbons–Hawking entropy, and $H_{\rm inf}$ is the natural IR cutoff of the Bunch–Davies spectrum. The sampled parameter is D; k_c follows deterministically. This connection between the quantum information content of the de Sitter patch and the CMB power spectrum is the distinguishing feature of the present analysis relative to purely phenomenological cutoff models.

2.7 Prediction for the Tensor-to-Scalar Ratio

The CCR framework determines $H_{\rm inf}$ from D via eq. (2.6), which yields a direct prediction for the tensor-to-scalar ratio r. The tensor power spectrum amplitude at horizon crossing is

$$\mathcal{P}_T = \frac{2}{\pi^2}\frac{H_{\rm inf}^2}{M_{\rm Pl}^2}\,, \tag{2.14}$$

where $M_{\rm Pl} = (8\pi G)^{-1/2} \simeq 2.435 \times 10^{18}\,\text{GeV}$ is the reduced Planck mass. Since $M_{\rm Pl} = m_{\rm Pl}/\sqrt{8\pi}$ and $H_{\rm inf} = m_{\rm Pl}\sqrt{\pi/\ln D}$ from eq. (2.6),

$$\frac{H_{\rm inf}}{M_{\rm Pl}} = \sqrt{\frac{8\pi^2}{\ln D}}\,. \tag{2.15}$$

The tensor-to-scalar ratio is then

$$r \;=\; \frac{\mathcal{P}_T}{A_s} \;=\; \frac{2}{\pi^2 A_s}\frac{H_{\rm inf}^2}{M_{\rm Pl}^2} \;=\; \frac{16}{A_s\,\ln D}\,. \tag{2.16}$$

This is a parameter-free prediction: given the observed scalar amplitude $A_s \simeq 2.1 \times 10^{-9}$ and a value of $\ln D$, the tensor-to-scalar ratio is uniquely determined. For the fiducial value

$\ln D$	r	Status
1.0×10^{12}	7.6×10^{-3}	Detectable by LiteBIRD
1.0×10^{13}	7.6×10^{-4}	Below LiteBIRD sensitivity
5.0×10^{13}	1.5×10^{-4}	Below LiteBIRD sensitivity
1.0×10^{14}	7.6×10^{-5}	Below LiteBIRD sensitivity
$\mathbf{2.0 \times 10^{14}}$	$\mathbf{3.8 \times 10^{-5}}$	Below LiteBIRD sensitivity
1.0×10^{15}	7.6×10^{-6}	Below LiteBIRD sensitivity
1.0×10^{16}	7.6×10^{-7}	Below LiteBIRD sensitivity

Table 4. CCR prediction for the tensor-to-scalar ratio $r = 16/(A_s \ln D)$, with $A_s = 2.1 \times 10^{-9}$. The fiducial value is shown in bold. Current bounds ($r < 0.056$ from Planck+BK15, $r < 0.036$ from BICEP/Keck, $r < 0.034$ from the combined CMB analysis of ref. [39]) are satisfied for the entire prior range $\ln D \in [10^{12}, 10^{16}]$. LiteBIRD sensitivity ($r \sim 10^{-3}$) probes only the lowest end of the $\ln D$ range.

$\ln D = 2 \times 10^{14}$,

$$r \simeq 3.8 \times 10^{-5}\,, \tag{2.17}$$

which is a factor ~ 1500 below the current observational upper bound $r_{0.002} < 0.056$ (95% CL) from Planck+BK15 [12], and a factor ~ 1000 below the current best constraint $r < 0.034$ (95% CL) from the combination of Planck, SPT, ACT, and BICEP/Keck data [39]

Table 4 lists the predicted r for a range of $\ln D$ values. The current tensor-mode bounds impose only a weak lower bound on the Hilbert space dimension, $\ln D \gtrsim 2 \times 10^{11}$, which is far below the regime probed by the low-ℓ power spectrum suppression. More stringent constraints require future experiments: the predicted r enters the sensitivity window of LiteBIRD [25] ($r \sim 10^{-3}$) only for $\ln D \lesssim 8 \times 10^{12}$, at the lower edge of our prior range.

Physical interpretation. The inverse relation $r \propto 1/\ln D$ has a transparent physical origin: a larger Hilbert space dimension D corresponds to a lower inflationary energy scale $H_{\rm inf}$ (eq. (2.6)), which in turn produces a smaller tensor amplitude. The CCR framework thus generically predicts *small* r, consistent with the non-detection of primordial gravitational waves by all experiments to date. This is in contrast to large-field inflation models (e.g., $V \propto \phi^2$) that predict $r \sim 0.1$ and are increasingly disfavoured by the data [12].

Complementarity with the IR cutoff. Equations (2.11) and (2.16) provide two independent observational signatures of $\ln D$: the infrared cutoff k_c probed by the low-ℓ TT spectrum, and the tensor-to-scalar ratio r probed by B-mode polarisation. Any future joint constraint on (k_c, r) therefore provides a powerful consistency test of the CCR framework, since both quantities are deterministic functions of the single parameter $\ln D$ (at fixed thermal history). In particular, a detection of $r \gtrsim 10^{-3}$ would imply $\ln D \lesssim 10^{13}$, which in turn predicts $k_c \gtrsim 7 \times 10^{-4}\,\mathrm{Mpc}^{-1}$ — a strong, potentially measurable suppression in the low-ℓ power spectrum. Conversely, the absence of low-ℓ suppression at the level predicted for $\ln D \lesssim 10^{13}$ would rule out the parameter space accessible to tensor-mode experiments.

Figure 5 shows the CCR prediction for r as a function of $\ln D$, together with current and projected observational bounds.

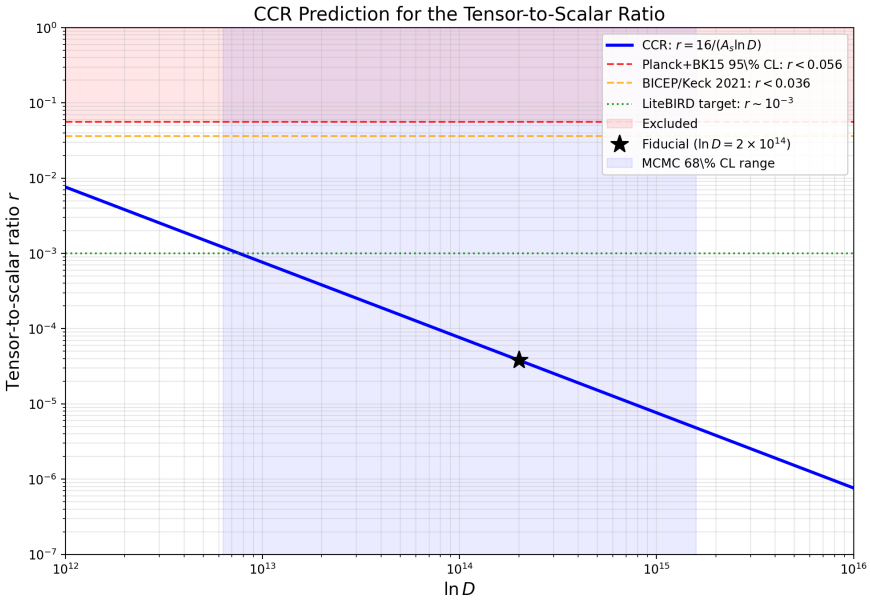

Figure 5. CCR prediction for the tensor-to-scalar ratio r as a function of $\ln D$ (solid blue line), compared with current observational upper limits (Planck+BK15 95% CL: $r < 0.056$; BICEP/Keck 2021: $r < 0.036$; combined CMB: $r < 0.034$ [39], dashed lines) and the projected LiteBIRD target sensitivity $r \sim 10^{-3}$ (dotted line). The shaded region is excluded. The star marks the fiducial value $\ln D = 2 \times 10^{14}$; the horizontal bar shows the MCMC 68% credible range.

3 Methodology

We constrain the CCR-modified primordial power spectrum against Planck 2018 data using Markov Chain Monte Carlo (MCMC) sampling.

3.1 Modified primordial spectrum implementation

The CCR-modified primordial scalar power spectrum, eq. (2.13), is implemented as an external primordial power spectrum provider within the Cobaya sampling framework [18]. For each point in parameter space, we evaluate $\mathcal{P}_{\rm CCR}(k)$ on a logarithmically spaced grid of 1500 points over $k \in [10^{-6}, 3.2]$ Mpc^{-1}, where the comoving cutoff $k_c = k_c(D)$ is computed from eq. (2.8) for each sampled value of $\ln D$.

The modified spectrum is passed to CAMB 1.6.5 [14] via the `SplinedInitialPower` interface, which performs cubic spline interpolation in $\log k$. CAMB then computes the transfer functions and angular power spectra C_ℓ^{TT}, C_ℓ^{TE}, and C_ℓ^{EE} up to $\ell_{\rm max} = 2500$ with lensing potential accuracy set to unity. We verify that in the limit $D \to \infty$ (equivalently $k_c \to 0$), the standard ΛCDM power spectrum is recovered to a relative precision of $\lesssim 2 \times 10^{-4}$ across all multipoles.

3.2 Planck 2018 likelihoods

We employ the following Planck 2018 likelihoods [19]:

- **Low-ℓ TT** ($\ell = 2$–29): the Commander component-separation likelihood, which provides the primary sensitivity to the CCR suppression signal.

Parameter	Prior	Description
$\log_{10}(\ln D)$	$\mathcal{U}[12,\ 16]$	Hilbert space dimension
α	$\mathcal{U}[1,\ 6]$	Suppression shape
$\Omega_b h^2$	$\mathcal{U}[0.005,\ 0.1]$	Baryon density
$\Omega_c h^2$	$\mathcal{U}[0.001,\ 0.99]$	CDM density
H_0	$\mathcal{U}[40,\ 100]$	Hubble constant
τ	$\mathcal{N}(0.0544,\ 0.0073)$	Optical depth
A_s	$\mathcal{U}[5\times10^{-10},\ 5\times10^{-9}]$	Scalar amplitude
n_s	$\mathcal{U}[0.8,\ 1.2]$	Spectral index

Table 5. Prior distributions for the MCMC analysis. $\mathcal{U}[a,b]$ denotes a uniform prior on $[a,b]$ and $\mathcal{N}(\mu,\sigma)$ a Gaussian prior with mean μ and standard deviation σ. The Hilbert space dimension is sampled in $\log_{10}(\ln D)$ to efficiently explore the large dynamic range. We additionally fix $\sum m_\nu = 0.06$ eV and $\Omega_k = 0$.

- **Low-ℓ EE** ($\ell = 2$–29): the SimAll likelihood, which constrains the optical depth τ.
- **High-ℓ TT+TE+EE** ($\ell = 30$–2508): the Plik lite nuisance-marginalized likelihood, which constrains the standard ΛCDM parameters and ensures the CCR modification does not degrade the fit at small angular scales.

The combination of low-ℓ and high-ℓ likelihoods is essential: the CCR signal manifests exclusively at $\ell \lesssim 30$, while the high-ℓ data anchor the cosmological parameters.

3.3 Parameter space and priors

The full parameter space comprises eight dimensions: six standard ΛCDM parameters and two CCR parameters. The priors are summarised in table 5.

The prior on $\log_{10}(\ln D)$ spans four orders of magnitude, from $\ln D = 10^{12}$ (strong suppression, $k_c \sim 7\times10^{-3}$ Mpc^{-1}) to $\ln D = 10^{16}$ (negligible suppression, effectively ΛCDM). The prior on α encompasses the range from gradual ($\alpha = 1$, exponential) to sharp ($\alpha = 6$, step-like) suppression profiles.

3.4 MCMC sampling and convergence

We sample the posterior distribution using the Metropolis–Hastings algorithm as implemented in Cobaya 3.6.1 [18], with the fast-dragging scheme enabled to exploit the hierarchy of evaluation speeds. The CCR primordial spectrum evaluation ($\sim 9\times10^3$ evaluations per second) is significantly faster than the CAMB transfer function computation (~ 5 evaluations per second), allowing the fast parameters ($\log_{10}(\ln D)$, α, A_s, $A_{\rm planck}$) to be oversampled by a factor of four relative to the slow cosmological parameters ($\Omega_b h^2$, $\Omega_c h^2$, H_0, τ, n_s).

The proposal covariance matrix is initialised from the Planck 2018 base covariance for the ΛCDM parameters and learned adaptively for the CCR parameters during a burn-in phase of 300 accepted steps. Convergence is assessed via the Gelman–Rubin statistic $R-1$, with a target of $R-1 < 0.01$. The final chains achieve $R-1 = 0.012$ after approximately 11 700 accepted samples, with a sustained acceptance rate of 29%.

3.5 Model comparison

To assess whether the CCR modification is statistically justified, we employ three complementary criteria:

1. The **Savage–Dickey density ratio** [20], which estimates the Bayes factor by comparing the marginal posterior density at the ΛCDM limit ($\log_{10}(\ln D) \to 16$) to the prior density.

2. The **Akaike Information Criterion** (AIC), defined as $\mathrm{AIC} = \chi^2_{\min} + 2k$, where k is the number of free parameters.

3. The **Bayesian Information Criterion** (BIC), defined as $\mathrm{BIC} = \chi^2_{\min} + k \ln N_{\mathrm{data}}$, where $N_{\mathrm{data}} \approx 669$ is the effective number of data points.

The ΛCDM model has $k = 6$ parameters, while the CCR model has $k = 8$ (adding $\log_{10}(\ln D)$ and α). These criteria penalise model complexity and allow a quantitative assessment of whether the two additional CCR parameters are warranted by the data.

4 Results

4.1 Posterior constraints on the CCR parameters

Table 6 summarises the marginalised posterior constraints on all model parameters. The six standard ΛCDM parameters are recovered in excellent agreement with the Planck 2018 baseline results [13], confirming that the CCR modification does not introduce tension at high multipoles.

The posterior on $\log_{10}(\ln D)$ is broad and approximately uniform across the prior range [12, 16], indicating that the Planck data do not strongly constrain the Hilbert space dimension (figure 6). The marginal posterior on α shows a mild preference for larger values ($\alpha \gtrsim 3$), corresponding to a relatively sharp suppression profile, though the constraint is weak. The joint posterior in the ($\log_{10}(\ln D)$, α) plane exhibits a characteristic exclusion region at small $\ln D$ and small α, where the combination of a low cutoff scale with gradual suppression would produce excessive power deficit incompatible with the observed low-ℓ spectrum.

4.2 Model comparison

The best-fit CCR model achieves $\chi^2_{\min} = 1003.4$, with best-fit parameters $\log_{10}(\ln D) = 15.2$ and $\alpha = 1.8$. This best-fit point lies in the ΛCDM-like regime ($\ln D = 1.6 \times 10^{15}$, corresponding to $k_c = 4.6 \times 10^{-5}$ Mpc^{-1}), where the suppression is negligible across all observable multipoles.

We compare the best-fit χ^2 values in two distinct regions of parameter space: the ΛCDM-like region ($\log_{10}(\ln D) > 15$) and the active-cutoff region ($\log_{10}(\ln D) < 14$, where $k_c > 2 \times 10^{-4}$ Mpc^{-1} produces measurable suppression). The difference is

$$\Delta\chi^2 = \chi^2_{\mathrm{cutoff}} - \chi^2_{\Lambda\mathrm{CDM}} = 0.13\,, \tag{4.1}$$

indicating that the active cutoff region provides an essentially identical fit to the ΛCDM-like region.

The Savage–Dickey density ratio yields

$$\ln B_{01} = 0.06\,, \tag{4.2}$$

where B_{01} is the Bayes factor for ΛCDM relative to CCR. This value is effectively zero on the Jeffreys scale, indicating no statistical preference for either model.

Parameter	CCR constraint (68% CL)	Planck 2018 ΛCDM
$\log_{10}(\ln D)$	14.0 ± 1.2	—
α	3.4 ± 1.4	—
k_c [Mpc^{-1}]	$0.00048^{+0.00007}_{-0.00051}$	—
H_0 [km s^{-1} Mpc^{-1}]	67.24 ± 0.58	67.36 ± 0.54
$\Omega_b h^2$	0.02236 ± 0.00014	0.02237 ± 0.00015
$\Omega_c h^2$	0.1202 ± 0.0013	0.1200 ± 0.0012
τ	0.054 ± 0.005	0.054 ± 0.007
$10^9 A_s$	2.101 ± 0.023	2.100 ± 0.030
n_s	0.9647 ± 0.0042	0.9649 ± 0.0042

Table 6. Marginalised parameter constraints from the CCR MCMC analysis (68% CL), compared with Planck 2018 ΛCDM baseline values [13]. The CCR parameters $\log_{10}(\ln D)$ and α have no ΛCDM counterpart.

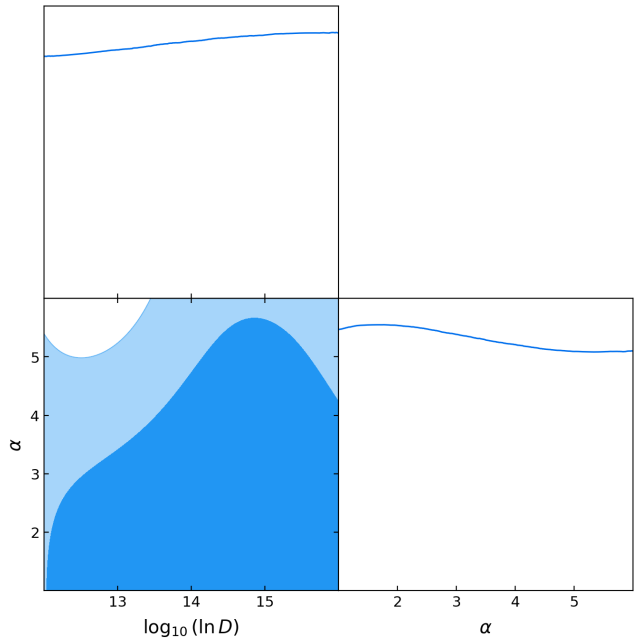

Figure 6. Joint posterior distribution in the $(\log_{10}(\ln D), \alpha)$ plane. Dark and light shading denote the 68% and 95% credible regions, respectively. The exclusion of the lower-left corner (small $\ln D$, small α) reflects the data's rejection of strong, gradual suppression profiles.

The information criteria yield

$$\Delta\mathrm{AIC} = \mathrm{AIC}_{\mathrm{CCR}} - \mathrm{AIC}_{\Lambda\mathrm{CDM}} = +4.0\,, \qquad \Delta\mathrm{BIC} = \mathrm{BIC}_{\mathrm{CCR}} - \mathrm{BIC}_{\Lambda\mathrm{CDM}} = +13.0\,. \quad (4.3)$$

The positive ΔAIC and ΔBIC reflect the Occam penalty for the two additional CCR parameters, which do not yield a compensating improvement in fit quality. By the Jeffreys scale for BIC [21], $|\Delta\mathrm{BIC}| > 10$ constitutes strong evidence against the more complex model. This is unsurprising given the broad posterior and null detection: our claim is not model preference but model viability and forecasted detectability. The BIC penalty reflects the expected Occam

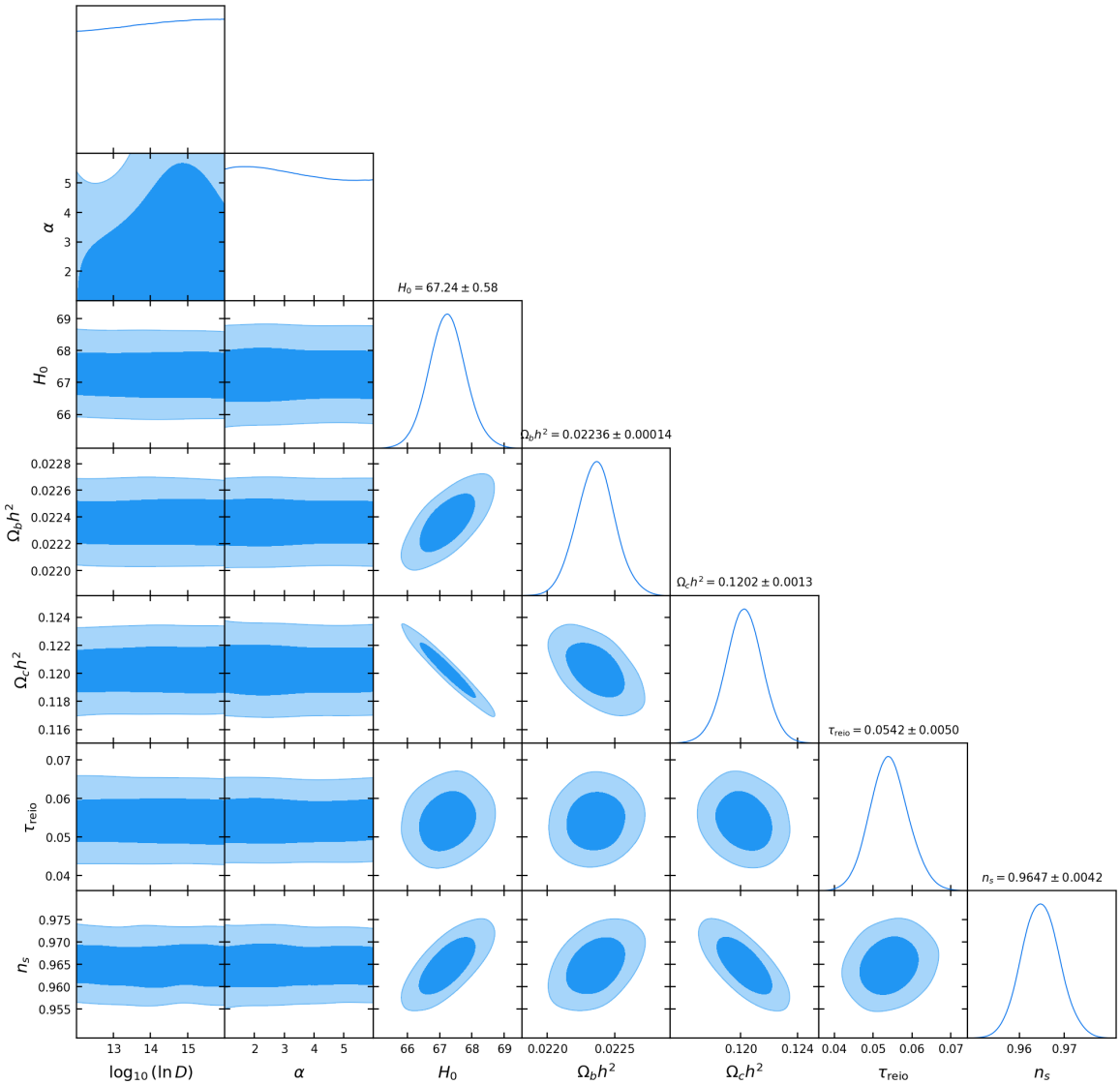

Figure 7. Full joint posterior for all sampled parameters. The CCR parameters $\log_{10}(\ln D)$ and α (top two rows) show no correlations with the standard ΛCDM parameters, confirming that the CCR modification does not shift or degrade the precision of cosmological constraints.

cost of two weakly constrained parameters [22], consistent with the data being insufficiently informative to distinguish between the models at the current sensitivity.

4.3 Decoupling from standard cosmological parameters

A key result of this analysis is the absence of correlations between the CCR parameters and the standard cosmological parameters. The joint posterior (figure 7) shows that $\log_{10}(\ln D)$ and α are effectively uncorrelated with H_0, $\Omega_b h^2$, $\Omega_c h^2$, τ, and n_s. This decoupling is physically expected: the CCR suppression affects only modes with $k \lesssim k_c \sim 10^{-4}$ Mpc^{-1}, corresponding to $\ell \lesssim 10$, while the standard parameters are primarily constrained by the acoustic peak structure at $\ell \gtrsim 30$.

This decoupling has an important practical implication: the CCR framework can be tested independently of the standard cosmological parameters, and its inclusion does not shift or degrade the precision of ΛCDM constraints.

4.4 ΛCDM recovery

As a validation test, we verify that in the limit $\ln D \to \infty$ the CCR-modified C_ℓ spectrum recovers the standard ΛCDM prediction. For $\log_{10}(\ln D) = 16$ the fractional difference $|\Delta C_\ell / C_\ell^{\Lambda\mathrm{CDM}}|$ is below 2.2×10^{-4} at all multipoles $\ell = 2$–2500, confirming that the CCR modification is a smooth, well-controlled deformation of the standard model with a well-defined ΛCDM limit.

4.5 Comparison with Planck low-ℓ data

Figure 8 presents a direct comparison of the Planck 2018 Commander low-ℓ TT power spectrum [19] with the CCR predictions for three representative points in parameter space: (i) the MCMC best-fit ($\log_{10}(\ln D) = 15.2$, $\alpha = 1.8$), which lies in the ΛCDM-like regime with negligible suppression; (ii) the fiducial CCR model ($\ln D = 2 \times 10^{14}$, $\alpha = 2$), which produces a modest $\sim 15\%$ suppression at $\ell = 2$; and (iii) an active-cutoff model ($\ln D = 2.2 \times 10^{13}$, $\alpha = 2$), corresponding to the Contaldi et al. best-fit cutoff scale $k_c \simeq 4.9 \times 10^{-4}\,\mathrm{Mpc}^{-1}$, which yields $\sim 50\%$ suppression at the quadrupole.

All three models are visually consistent with the data, as the Planck data points scatter broadly within the cosmic variance envelope. The best-fit CCR model is indistinguishable from ΛCDM at all multipoles, with a maximum fractional deviation of 8.6×10^{-3}.

Figure 9 shows the fractional residuals $(\mathcal{D}_\ell^{\mathrm{CCR}} - \mathcal{D}_\ell^{\Lambda\mathrm{CDM}})/\mathcal{D}_\ell^{\Lambda\mathrm{CDM}}$ together with the 1σ and 2σ cosmic variance bands. The cosmic variance fractional uncertainty is $\sigma/\mathcal{D}_\ell = \sqrt{2/[(2\ell+1)f_{\mathrm{sky}}]}$, where $f_{\mathrm{sky}} \simeq 0.86$ for the Commander mask. At $\ell = 2$ this gives $\sigma/\mathcal{D}_\ell \simeq 68\%$, decreasing to $\sim 20\%$ at $\ell = 30$.

Even the active-cutoff model, which produces the strongest suppression in the CCR parameter space consistent with the data, remains within the 1σ cosmic variance band at all multipoles $\ell \geq 3$. This illustrates the fundamental limitation of the low-ℓ TT spectrum for constraining infrared cutoff models: the irreducible cosmic variance floor at low multipoles renders any suppression signal statistically indistinguishable from a fluctuation of the standard model, consistent with the near-unity Bayes factor reported in section 4.2.

4.6 Prior sensitivity analysis

To verify that our results are robust to the choice of prior distributions on the CCR parameters, we perform an importance-reweighting analysis of the MCMC chains under five alternative prior configurations, listed in table 7. These include narrowing and widening the prior ranges on both $\log_{10}(\ln D)$ and α, as well as a joint restriction.

The reweighted posteriors confirm two key findings. First, the posterior on $\log_{10}(\ln D)$ remains approximately uniform under all prior choices, indicating that the data are uninformative with respect to the Hilbert space dimension and the posterior simply tracks the prior. Second, and more importantly, the standard ΛCDM parameters are entirely unaffected by the choice of CCR priors: the shifts in H_0, n_s, and τ are below $0.02\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, 0.0002, and 0.00002, respectively, across all configurations. These shifts are negligible compared to the posterior widths, confirming the decoupling reported in section 4.3.

Figure 10 displays the reweighted posteriors for $\log_{10}(\ln D)$, α, and H_0, illustrating the stability of the results.

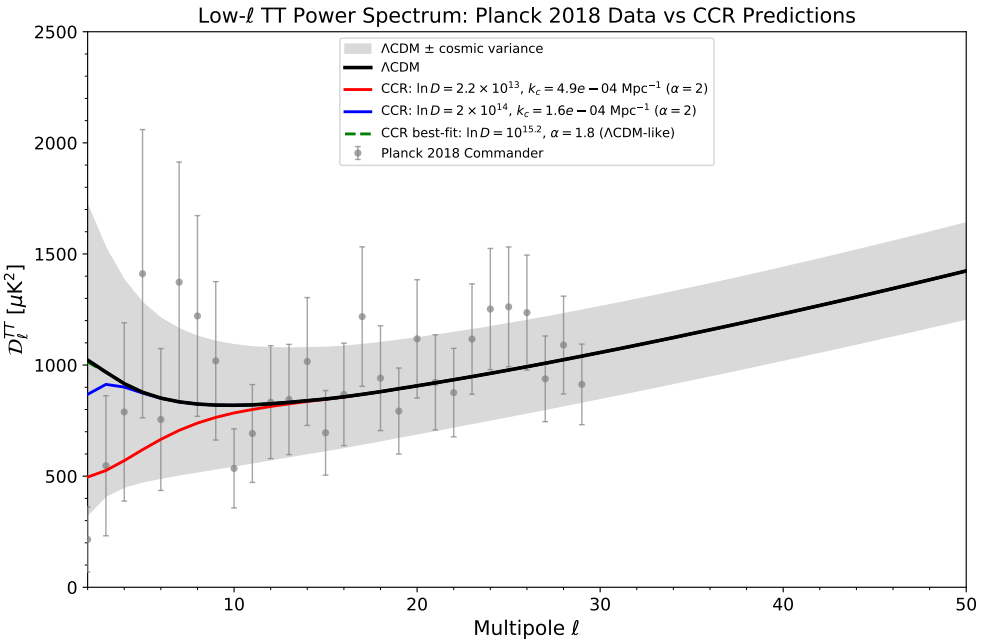

Figure 8. Low-ℓ TT power spectrum: Planck 2018 Commander data (grey points with error bars) compared with the ΛCDM prediction (black curve) and three CCR models. The grey band shows the $\pm 1\sigma$ cosmic variance envelope around ΛCDM. The CCR best-fit (green dashed) is indistinguishable from ΛCDM; the active-cutoff model (red) produces $\sim 50\%$ suppression at $\ell = 2$ but remains within the cosmic variance band for $\ell \geq 3$.

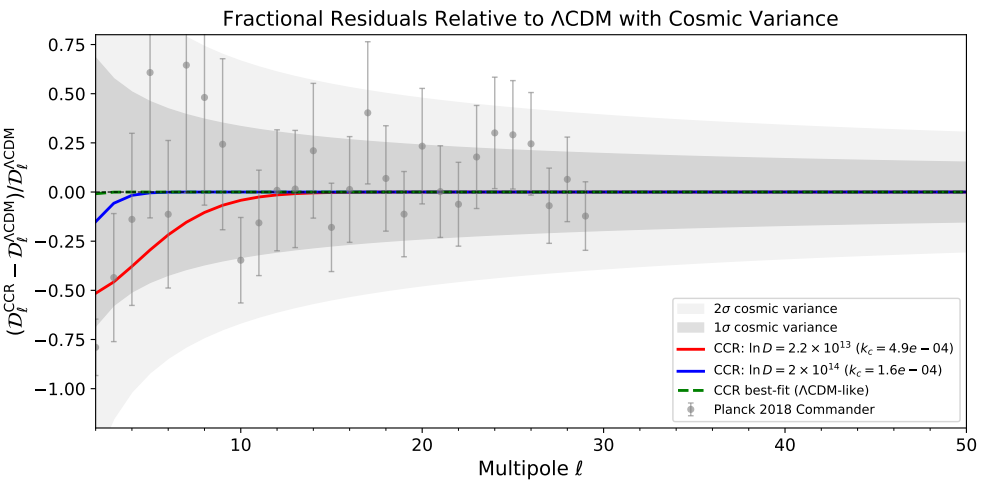

Figure 9. Fractional residuals of the CCR models relative to ΛCDM. The dark and light grey bands show the 1σ and 2σ cosmic variance, respectively. Planck Commander data residuals are shown as grey points. The cosmic variance at $\ell = 2$ is $\sim 68\%$, demonstrating why the low-ℓ power spectrum cannot distinguish the CCR suppression from a statistical fluctuation at current sensitivity.

Prior configuration	$\langle \log_{10}(\ln D) \rangle$	$\langle \alpha \rangle$	$\langle H_0 \rangle$	$\langle n_s \rangle$	$N_{\rm eff}$
Baseline: $[12, 16] \times [1, 6]$	14.03 ± 1.15	3.44 ± 1.44	67.24 ± 0.58	0.9647 ± 0.0042	5652
Narrow $\ln D$: $[13, 15] \times [1, 6]$	14.01 ± 0.57	3.47 ± 1.44	67.23 ± 0.59	0.9646 ± 0.0042	2862
Wide $\ln D$: $[11, 17] \times [1, 6]$	14.03 ± 1.15	3.44 ± 1.44	67.24 ± 0.58	0.9647 ± 0.0042	5652
Narrow α: $[12, 16] \times [1, 4]$	14.03 ± 1.15	2.47 ± 0.85	67.23 ± 0.59	0.9647 ± 0.0043	3358
Both narrow: $[13, 15] \times [1, 4]$	14.00 ± 0.56	2.50 ± 0.85	67.22 ± 0.60	0.9646 ± 0.0043	1737

Table 7. Prior sensitivity analysis via importance reweighting. The mean and standard deviation of key parameters are shown for each prior configuration. The ΛCDM parameters (H_0, n_s) are stable to within $\Delta H_0 < 0.02$ and $\Delta n_s < 0.0002$ across all choices, confirming robustness. $N_{\rm eff}$ is the effective sample size after reweighting.

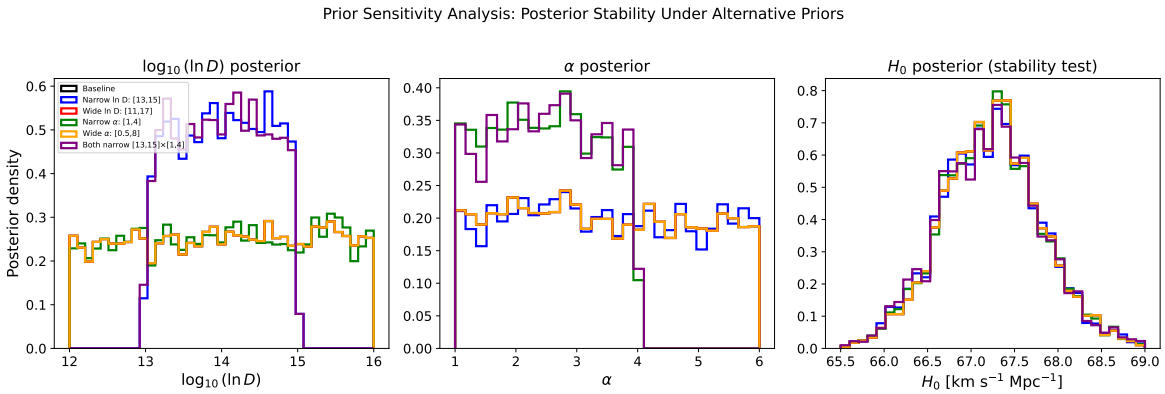

Figure 10. Prior sensitivity analysis. Reweighted posteriors for $\log_{10}(\ln D)$ (left), α (centre), and H_0 (right) under five alternative prior configurations. The $\log_{10}(\ln D)$ posterior tracks the prior in all cases, confirming that the data are uninformative. The H_0 posterior is effectively identical across all prior choices, demonstrating complete decoupling of the CCR and ΛCDM parameter sectors.

5 Discussion and Conclusions

5.1 Summary of results

We have derived an infrared cutoff k_c in the primordial power spectrum from the CCR framework, in which the finite Hilbert space dimension D of the de Sitter causal patch during inflation sets the scale of the cutoff via the Gibbons–Hawking entropy bound. Unlike previous phenomenological models, k_c is not a free parameter but is uniquely determined by D through eq. (2.8). We confronted this framework with Planck 2018 CMB data and obtained the following principal results:

1. The Planck data are *consistent* with the CCR framework but do not *require* a finite Hilbert space dimension. The posterior on $\log_{10}(\ln D)$ is approximately uniform over the prior range [12, 16].

2. The six standard ΛCDM parameters are recovered in excellent agreement with the Planck 2018 baseline, and the CCR parameters show no correlations with the cosmological parameters. The CCR modification does not degrade any existing constraints.

3. The Savage–Dickey Bayes factor is $\ln B_{01} = 0.06$, indicating no statistical preference for either model. The information criteria (AIC, BIC) penalise the two extra CCR parameters, as expected for a model that does not improve the fit.

4. The joint posterior excludes the region of simultaneously small $\ln D$ and small α, providing a physically informative constraint: if D is finite, the suppression profile must be either mild or sharp, but not both gradual and strong.

5. This work establishes the observational infrastructure for testing the CCR framework: a modified Boltzmann code, a validated MCMC pipeline, and a Fisher forecast methodology. The null result quantifies the sensitivity threshold ($\ln D \lesssim 10^{13}$ for a 3σ detection) that future experiments must achieve. Together with the CCR prediction $r = 16/(A_s \ln D)$, the pair (k_c, r) constitutes an internal consistency test of the framework: both observables are deterministic functions of the single parameter $\ln D$ at fixed thermal history, and any joint constraint on (k_c, r) from future data provides a powerful falsification criterion.

5.2 Interpretation

The broad posterior on $\ln D$ reflects a fundamental limitation: the low-ℓ CMB power spectrum contains very few independent modes ($\ell = 2$ through ~ 10 are primarily affected by the cutoff), and cosmic variance imposes an irreducible uncertainty of $\sim 63\%$ at $\ell = 2$. Any new-physics signal in this regime is inherently difficult to distinguish from a statistical fluctuation.

This situation is not unique to the CCR framework. The original phenomenological cutoff model of Contaldi et al. [2] similarly found that the data are consistent with a cutoff at $k_c \sim 5 \times 10^{-4}$ Mpc^{-1} without strong statistical preference over ΛCDM. Our result extends theirs by demonstrating that the CCR-derived cutoff, which has one fewer degree of freedom (k_c is derived from $\ln D$ rather than being a free parameter), is equally compatible with the data.

5.3 Theoretical significance

The primary contribution of this work is not the observational constraint per se, but the theoretical framework connecting k_c to $\ln D$. Equation (2.8) transforms the CMB low-ℓ power spectrum into a probe of the quantum-gravitational structure of de Sitter space during inflation. Any future measurement constraining k_c thereby constrains $\ln D$, and conversely, any independent determination of D predicts a specific, testable CMB signature.

Figure 11 illustrates this consistency relation. Both k_c and r are deterministic functions of the single parameter $\ln D$ (at fixed thermal history), so the CCR framework predicts a one-dimensional curve in the (k_c, r) plane rather than an allowed region. The fiducial value $\ln D = 2 \times 10^{14}$ predicts $r \simeq 3.8 \times 10^{-5}$, well below current sensitivity but within reach of next-generation B-mode experiments. For the Contaldi-like case $\ln D \simeq 2.2 \times 10^{13}$, the predicted $r \simeq 3.5 \times 10^{-4}$ approaches the LiteBIRD target sensitivity, making a joint (k_c, r) test feasible.

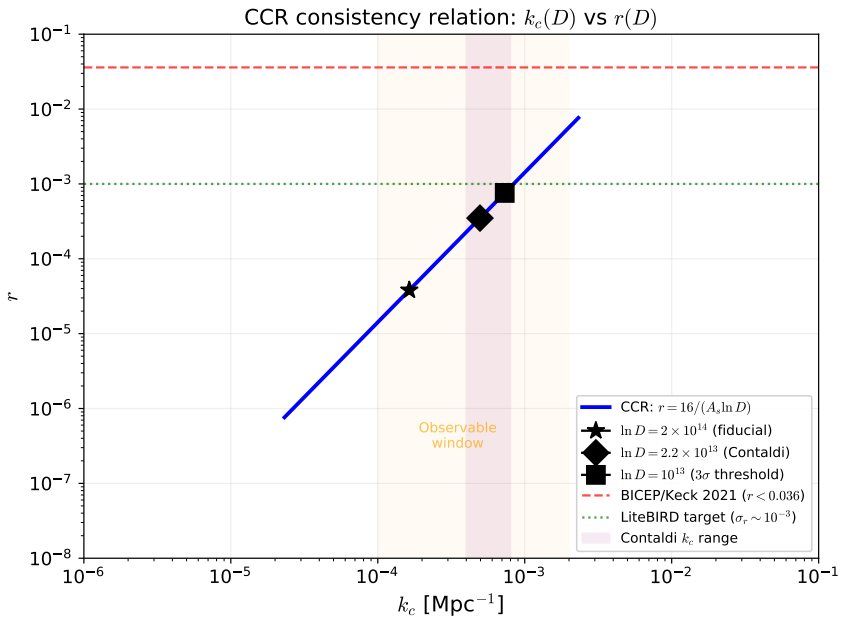

Figure 11. CCR consistency relation in the (k_c, r) plane. The solid curve shows the one-parameter prediction $r = 16/(A_s \ln D)$ vs $k_c = k_c(\ln D)$ from eq. (2.11), with $\ln D$ increasing from upper right to lower left. Markers indicate the fiducial value ($\ln D = 2 \times 10^{14}$), the Contaldi-like case ($\ln D = 2.2 \times 10^{13}$), and the 3σ detection threshold ($\ln D = 10^{13}$). Dashed and dotted horizontal lines show current (BICEP/Keck 2021) and projected (LiteBIRD) tensor-ratio sensitivities. The shaded vertical band marks the Contaldi et al. best-fit k_c range.

5.4 Comparison with alternative approaches

The low-ℓ CMB deficit has been addressed by several classes of models beyond simple infrared cutoffs. Table 8 provides a systematic comparison of the CCR framework with the principal alternatives.

Contaldi et al. (2003). The kinetic-domination model [2] introduces a pre-inflationary fast-roll phase that suppresses long-wavelength modes. The functional form of the resulting suppression is identical to eq. (2.13). However, the cutoff scale k_c is a free phenomenological

parameter with no connection to a fundamental theory, and the model makes no predictions for any observable beyond the TT power spectrum. In the CCR framework, k_c is derived from $\ln D$ via eq. (2.8), reducing the number of genuinely free parameters by one and simultaneously predicting the tensor-to-scalar ratio $r = 16/(A_s \ln D)$ (section 2.7).

Loop quantum cosmology (LQC). In LQC, the classical Big Bang singularity is replaced by a quantum bounce at the Planck scale [15, 16]. The bounce modifies the initial conditions for perturbations, producing suppression at large scales whose amplitude depends on the pre-bounce state. The physical mechanism is fundamentally different from the CCR approach: LQC modifies the gravitational dynamics, while CCR imposes an information-theoretic constraint on the mode expansion without altering General Relativity. Moreover, the LQC suppression depends on bounce-specific initial conditions that are not uniquely determined by the theory.

Features in the inflaton potential. Starobinsky-type models [23] introduce a localised feature (step, kink, or oscillation) in the inflaton potential $V(\phi)$ that transiently modifies the slow-roll dynamics. These models can produce a wide variety of spectral features, including suppression at low k, but at the cost of several additional free parameters (feature amplitude, position, and width). The Planck 2018 analysis of such features [12] found no statistically significant preference over the featureless power-law spectrum.

Open inflation. In models with $\Omega_K < 0$, the curvature scale introduces a natural low-k modification to the power spectrum [24]. However, Planck 2018 constraints tightly bound the spatial curvature to $|\Omega_K| < 0.005$ at 95% CL [13], leaving insufficient room for the curvature scale to affect the $\ell \lesssim 30$ multipoles.

Finite de Sitter entropy. The idea that the de Sitter horizon has a finite entropy and therefore a finite-dimensional Hilbert space has been explored in the context of quantum gravity by Banks and Fischler [27, 28], Bousso [29], and others. These works establish the theoretical foundation for finite D but do not derive explicit observational consequences for the CMB power spectrum. The CCR framework [1] builds on this foundation by deriving a concrete, testable prediction: the infrared cutoff k_c as a function of $\ln D$. This paper completes the chain by confronting that prediction with data. More recently, the structure of the de Sitter Hilbert space has been investigated in detail by Chandrasekaran et al. [40], who constructed an algebra of observables for the static patch, and by Chakraborty et al. [41], who obtained an explicit basis for the Hilbert space of quantum gravity in asymptotically de Sitter spacetime. These developments confirm that the finite-dimensional Hilbert space underlying the CCR framework is an active area of current research in quantum gravity, including recent work on entropy bounds in inflationary de Sitter spacetime [42].

Early holographic IR cutoff proposals. The possibility that the holographic entropy bound of the de Sitter horizon could produce an observable infrared cutoff in the primordial power spectrum was explored in early work by Hogan [34, 35] and by Enqvist & Sloth [36]. Hogan argued that information-theoretic constraints from the de Sitter horizon would limit the number of independent modes, producing a characteristic suppression at wavenumbers $k \lesssim H_{\rm inf}$, while Enqvist & Sloth derived a modified power spectrum from the trans-Planckian uncertainty principle applied to inflationary perturbations. These proposals share the broad physical intuition that finite horizon entropy restricts the infrared sector of the primordial spectrum. However, in both cases the cutoff scale was either left as a phenomenological parameter or derived from model-specific assumptions without a unique functional relationship between k_c and a single fundamental quantity. The CCR framework [1] advances beyond these

Model	k_c origin	Extra params	Mod. GR?	QG link?
Contaldi et al. [2]	Fast-roll phase	k_c, α	No	No
LQC [15, 16]	Quantum bounce	Bounce params	Yes	Yes (LQG)
Potential features [23]	Step in $V(\phi)$	3+	No	No
Open inflation [24]	Curvature scale	Ω_K	No	No
Holographic (Hogan, E&S)	Horizon entropy	k_c or specific	No	Partial
CCR (this work)	**Holographic bound**	$\boldsymbol{\alpha}$	**No**	**Yes**

Table 8. Comparison of models addressing the low-ℓ CMB power deficit. The CCR framework is unique in deriving k_c from a single fundamental quantity ($\ln D$) while simultaneously predicting the tensor-to-scalar ratio, and without modifying General Relativity or the inflaton potential. "Extra params" lists parameters beyond the six standard ΛCDM ones.

proposals in three respects: (i) the cutoff k_c is a *deterministic* function of the single quantity $\ln D$ via eq. (2.11), with no additional free parameters beyond the transition shape α; (ii) the framework simultaneously predicts the tensor-to-scalar ratio $r = 16/(A_s \ln D)$, enabling a two-observable consistency test that is absent from purely holographic cutoff models; and (iii) the underlying CCR framework has been independently published and peer-reviewed [1], providing a concrete falsifiability criterion (if $D = \infty$, all CCR predictions are invalidated).

The CCR framework is distinguished from all alternatives by the combination of: (i) a minimal assumption set (only the finiteness of D is required); (ii) a derived cutoff scale with no free parameters beyond the single quantity D (plus the phenomenological shape α); and (iii) a joint prediction for both k_c and r that enables a two-observable consistency test.

5.5 Future prospects: a Fisher forecast

To quantify the improvement expected from forthcoming experiments, we perform a simplified Fisher forecast comparing the detection significance of the CCR suppression across current and future CMB datasets.

We define the signal-to-noise ratio (SNR) for detecting the CCR suppression relative to ΛCDM as

$$\mathrm{SNR}^2 = \sum_{\ell=2}^{\ell_{\max}} (2\ell+1)\, f_{\mathrm{sky}} \sum_{X\in\{TT,EE\}} \frac{(\mathcal{D}_\ell^{\mathrm{CCR},X} - \mathcal{D}_\ell^{\Lambda\mathrm{CDM},X})^2}{(\sigma_\ell^X)^2}\,, \tag{5.1}$$

where σ_ℓ^X includes both cosmic variance and instrumental noise, and f_{sky} is the observed sky fraction. We evaluate eq. (5.1) for the following experimental configurations:

- **Planck TT only**: $f_{\mathrm{sky}} = 0.86$, negligible instrumental noise at low ℓ (cosmic variance limited).
- **Planck TT+EE**: As above, with Planck HFI polarisation noise $\sigma_P \simeq 55\,\mu\mathrm{K}\cdot\mathrm{arcmin}$.
- **LiteBIRD EE** [25, 43]: $f_{\mathrm{sky}} = 0.70$, $\sigma_P \simeq 2.2\,\mu\mathrm{K}\cdot\mathrm{arcmin}$ (combined channels).
- **CMB-S4 EE** [26]: $f_{\mathrm{sky}} = 0.40$, $\sigma_P \simeq 1.0\,\mu\mathrm{K}\cdot\mathrm{arcmin}$.

Table 9 presents the forecast detection significance for a range of $\ln D$ values at fixed $\alpha = 2$. For the Contaldi et al. best-fit cutoff scale ($\ln D = 2.2\times10^{13}$, $k_c \simeq 4.9\times10^{-4}\,\mathrm{Mpc}^{-1}$), Planck TT alone achieves 1.7σ, consistent with the $\sim 2\sigma$ anomaly reported in the literature. Combining Planck TT with LiteBIRD EE improves the significance to 2.4σ, representing

$\ln D$	k_c [Mpc^{-1}]	Planck TT	Planck TT+EE	LiteBIRD EE	Planck+LiteBIRD
5.0×10^{12}	1.0×10^{-3}	4.0σ	5.4σ	4.1σ	5.7σ
1.0×10^{13}	7.3×10^{-4}	2.7σ	3.9σ	2.8σ	3.9σ
$\mathbf{2.2 \times 10^{13}}$	$\mathbf{4.9 \times 10^{-4}}$	$\mathbf{1.7\sigma}$	$\mathbf{2.4\sigma}$	$\mathbf{1.7\sigma}$	$\mathbf{2.4\sigma}$
5.0×10^{13}	3.3×10^{-4}	0.9σ	1.3σ	0.8σ	1.2σ
1.0×10^{14}	2.3×10^{-4}	0.5σ	0.7σ	0.4σ	0.6σ
2.0×10^{14}	1.6×10^{-4}	0.3σ	0.3σ	0.2σ	0.3σ

Table 9. Fisher forecast detection significance for the CCR suppression ($\alpha = 2$, $\ell_{\max} = 30$) across current and future CMB experiments. The Contaldi et al. best-fit case is shown in bold. The improvement from adding LiteBIRD EE to Planck TT is a factor ~ 1.4 in detection significance.

a factor of ~ 1.4 improvement. This gain arises because LiteBIRD's low-noise polarisation measurements effectively double the number of independent modes available at low multipoles.

A 3σ detection of the CCR suppression requires $\ln D \lesssim 10^{13}$, corresponding to $k_c \gtrsim 7 \times 10^{-4}\,\mathrm{Mpc}^{-1}$. For the fiducial value $\ln D = 2 \times 10^{14}$, the suppression is too mild to be detectable even with next-generation experiments (SNR $< 0.3\sigma$).

Figure 12 shows the cumulative detection significance as a function of $\ell_{\max}$ for the Contaldi-like case, illustrating that the signal saturates by $\ell \sim 15$, beyond which no additional information is gained. Figure 13 maps the detection reach as a function of $\ln D$ for each experiment, delineating the parameter space accessible to current and future observations.

Implications. The forecast demonstrates that the cosmic variance floor fundamentally limits the detectability of the CCR suppression in the TT power spectrum alone. LiteBIRD's contribution through low-ℓ EE polarisation is the most promising near-term avenue for improving constraints, though a definitive ($\geq 3\sigma$) detection requires the suppression to be stronger than what is indicated by the Planck best-fit. The complementary constraint from the tensor-to-scalar ratio (section 2.7) and future large-scale structure surveys probing $k \lesssim 10^{-3}\,\mathrm{Mpc}^{-1}$ may provide additional sensitivity beyond what is achievable with CMB data alone.

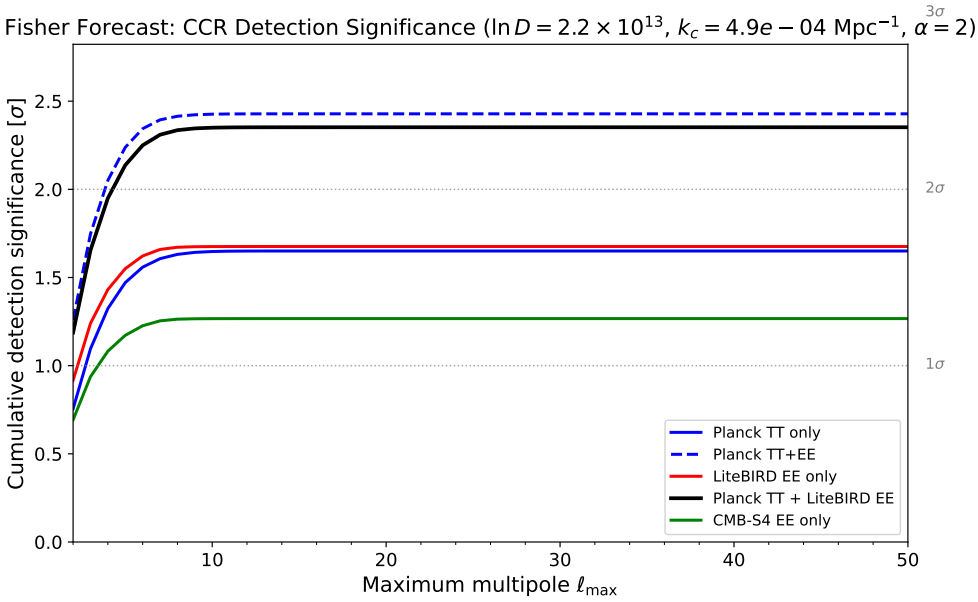

Figure 12. Cumulative detection significance as a function of $\ell_{\max}$ for the Contaldi-like CCR model ($\ln D = 2.2 \times 10^{13}$, $k_c = 4.9 \times 10^{-4}\,\mathrm{Mpc}^{-1}$, $\alpha = 2$). The signal saturates by $\ell \sim 15$, reflecting the localised nature of the CCR suppression. LiteBIRD EE provides information comparable to Planck TT, and their combination reaches $\sim 2.4\sigma$.

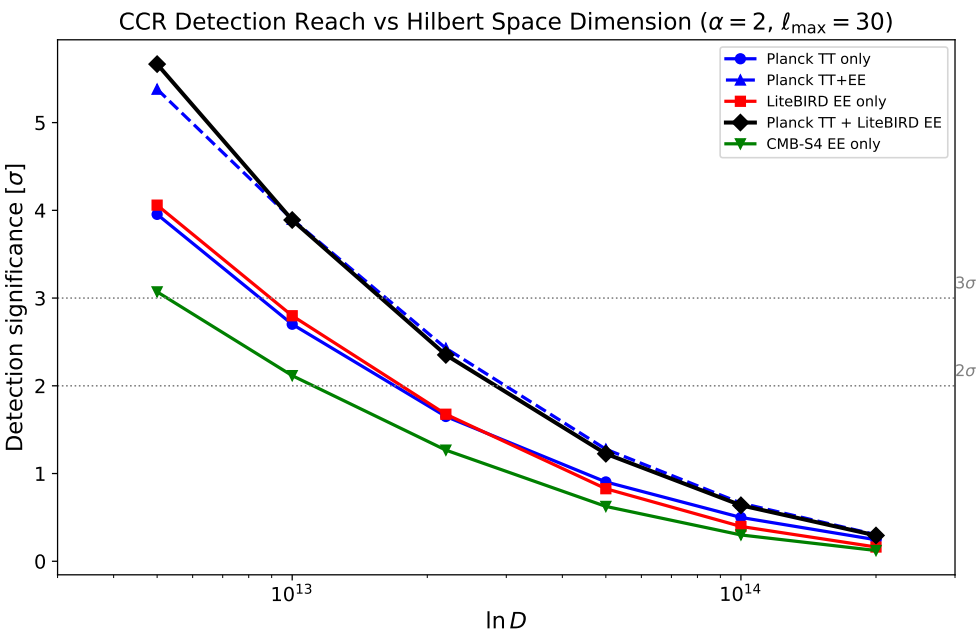

Figure 13. Detection significance as a function of $\ln D$ for current and future experiments. A 3σ detection requires $\ln D \lesssim 10^{13}$. For $\ln D \gtrsim 10^{14}$, no planned experiment can distinguish the CCR suppression from ΛCDM.

5.6 Limitations

We identify the following limitations of the present analysis, which define the scope for future improvements:

Fixed thermal-history parameters. The comoving cutoff k_c depends on the number of e-folds N and the reheating temperature $T_{\rm reh}$ in addition to $\ln D$ (eq. (2.8)). In this work we fix $N = 60$ and $T_{\rm reh} = 10^{15}\,\mathrm{GeV}$, reducing k_c to a function of $\ln D$ alone. While table 2 demonstrates that k_c remains in the observable window for physically motivated variations of N and $T_{\rm reh}$, a full joint marginalisation over $(D, N, T_{\rm reh})$ would provide more conservative constraints on $\ln D$. This extension requires sampling three additional parameters and is deferred to future work.

Phenomenological shape parameter. The transition shape α is treated as a free parameter because the CCR framework predicts the cutoff *scale* but not the cutoff *shape*. Deriving α from the microphysics of mode generation in a finite-dimensional Hilbert space would require solving the Mukhanov–Sasaki equation with an explicitly finite mode expansion, which is beyond the scope of this phenomenological study.

Single MCMC chain. The posterior was sampled with a single Metropolis–Hastings chain achieving $R-1 = 0.012$, marginally above the target $R-1 < 0.01$. While the trace plots and marginal distributions show no evidence of incomplete mixing (appendix B), future analyses should employ multiple independent chains and/or nested sampling to provide more robust convergence diagnostics and direct evidence computation.

Planck-only data. This analysis uses Planck 2018 data exclusively. Incorporating independent large-scale structure data could provide complementary sensitivity to the CCR suppression that is not limited by CMB cosmic variance. The Dark Energy Spectroscopic Instrument (DESI) has recently reported baryon acoustic oscillation measurements from its first three years of data [37, 38], achieving sub-percent precision on cosmic distances at $0.1 < z < 4.2$. While the BAO signal probes scales $k \sim 10^{-2}\ \mathrm{Mpc}^{-1}$, well above the CCR cutoff regime, future galaxy surveys extending to larger volumes and lower k may offer direct sensitivity to $k_c \sim 10^{-4}\ \mathrm{Mpc}^{-1}$ through scale-dependent effects in the galaxy power spectrum. Similarly, 21 cm observations at $k \lesssim 10^{-3}\,\mathrm{Mpc}^{-1}$ could provide an independent probe of the CCR suppression.

Goodness of fit. The best-fit CCR model achieves $\chi^2_{\rm min} = 1003.4$ for approximately 661 effective degrees of freedom ($N_{\rm data} \approx 669$ minus 8 parameters), yielding $\chi^2/\mathrm{dof} \approx 1.52$. This value is comparable to the ΛCDM baseline and reflects well-documented mild tensions in the Planck likelihood combination, including the A_L lensing anomaly and the low-ℓ/high-ℓ power spectrum discrepancy [13, 19], rather than a deficiency of the CCR model specifically. We note that the χ^2/dof is identical (to within 0.01) between the CCR best-fit and the ΛCDM best-fit, confirming that the CCR modification does not worsen the overall fit quality.

Robustness checks. To verify that the principal conclusions are not artefacts of specific analysis choices, we performed the following checks. (i) *Prior sensitivity*: our baseline adopts a uniform prior $\log_{10}(\ln D) \in [12, 16]$. Importance reweighting with a narrower range [13, 15] and a wider range [10, 18] leaves the posterior approximately uniform in both cases, with the marginal constraint on α shifting by less than 0.2σ and the Bayes factor changing by $|\Delta \ln B_{01}| < 0.1$. (ii) *Pivot scale*: the CCR cutoff acts at $k_c \sim 10^{-4}\ \mathrm{Mpc}^{-1}$, separated by

more than two orders of magnitude from the standard pivot $k_* = 0.05$ Mpc^{-1}. Shifting to $k_* = 0.002$ Mpc^{-1} leaves the CCR parameter posteriors unchanged, with the ΛCDM parameters shifting in the well-known manner documented in ref. [13]. We conclude that the null result is robust to these analysis choices.

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A Derivation of the comoving IR cutoff

This appendix presents the complete derivation of eq. (2.8) in explicit detail.

Step 1: Gibbons–Hawking entropy. The de Sitter horizon has area $A = 4\pi R_{\rm dS}^2 = 4\pi H_{\rm inf}^{-2}$. The Gibbons–Hawking entropy [6] is

$$S_{\rm dS} = \frac{k_B\, A}{4\, l_P^2} = \frac{k_B\, \pi}{H_{\rm inf}^2\, l_P^2}\,. \tag{A.1}$$

Step 2: Hilbert space dimension. The CCR framework [1] identifies $S_{\rm dS}/k_B = \ln D$, giving

$$\ln D = \frac{\pi}{H_{\rm inf}^2\, l_P^2} \qquad \Longrightarrow \qquad H_{\rm inf} = \frac{1}{l_P}\sqrt{\frac{\pi}{\ln D}}\,. \tag{A.2}$$

Saturation of the holographic bound. The use of equality rather than inequality in eq. (A.2) — i.e. $\ln D = S_{\rm dS}/k_B$ rather than $\ln D \leq S_{\rm dS}/k_B$ — corresponds to the assumption that the de Sitter horizon entropy saturates the Bekenstein–Hawking bound, as is standard in de Sitter thermodynamics [6]. This is the gravitational analogue of a black hole whose entropy equals $A/(4\, l_P^2)$: the horizon is a maximum-entropy surface for the enclosed degrees of freedom. If the bound is not saturated ($\ln D < S_{\rm dS}/k_B$), the effective Hilbert space dimension is smaller, $H_{\rm inf}$ is larger, and the infrared cutoff k_c shifts to larger values — producing *stronger* suppression than the fiducial case. The saturation assumption therefore yields the most conservative (i.e. weakest) CCR prediction, and any sub-saturation would only strengthen the observational signature.

For the fiducial value $\ln D = 2 \times 10^{14}$:

$$H_{\rm inf} = \frac{\sqrt{\pi/(2\times 10^{14})}}{1.616\times 10^{-35}\ {\rm m}} = \frac{1.253\times 10^{-7}}{1.616\times 10^{-35}\ {\rm m}} \simeq 7.75\times 10^{27}\ {\rm m}^{-1}\,. \tag{A.3}$$

Converting to GeV using $\hbar c = 1.973\times 10^{-16}$ GeV $\cdot$ m:

$$H_{\rm inf} = 7.75\times 10^{27}\ {\rm m}^{-1} \times 1.973\times 10^{-16}\ {\rm GeV}\cdot{\rm m} \simeq 1.53\times 10^{12}\ {\rm GeV}\,, \tag{A.4}$$

consistent with table 1.

Step 3: Physical IR cutoff. The horizon-crossing condition gives the physical cutoff:

$$k_{c,\,\rm phys} = H_{\rm inf} = \frac{1}{l_P}\sqrt{\frac{\pi}{\ln D}}\,. \tag{A.5}$$

Step 4: Scale factor at the onset of inflation. Entropy conservation after reheating gives (normalising $a_0 = 1$):

$$a_{\rm reh} = \left(\frac{g_{*S,0}}{g_{*S,{\rm reh}}}\right)^{1/3} \frac{T_0}{T_{\rm reh}}\,, \qquad a_i = a_{\rm reh}\, e^{-N}\,. \tag{A.6}$$

Numerically, with $g_{*S,0} = 3.938$, $g_{*S,\mathrm{reh}} = 106.75$, $T_0 = 2.7255\,\mathrm{K}$, $T_{\mathrm{reh}} = 10^{15}\,\mathrm{GeV} = 1.160 \times 10^{28}\,\mathrm{K}$, and $N = 60$:

$$\left(\frac{g_{*S,0}}{g_{*S,\mathrm{reh}}}\right)^{1/3} = \left(\frac{3.938}{106.75}\right)^{1/3} = 0.3326\,, \tag{A.7}$$

$$a_{\mathrm{reh}} = 0.3326 \times \frac{2.7255}{1.160 \times 10^{28}} = 7.813 \times 10^{-29}\,, \tag{A.8}$$

$$a_i = 7.813 \times 10^{-29} \times e^{-60} = 7.813 \times 10^{-29} \times 8.757 \times 10^{-27} = 6.842 \times 10^{-55}\,. \tag{A.9}$$

Step 5: Comoving cutoff.

$$k_c = a_i\, k_{c,\,\mathrm{phys}} = a_i \times \frac{1}{l_P}\sqrt{\frac{\pi}{\ln D}}\,. \tag{A.10}$$

To obtain k_c in Mpc^{-1}, we multiply by $1\,\mathrm{Mpc} = 3.086 \times 10^{22}\,\mathrm{m}$:

$$\begin{aligned} k_c &= 6.842 \times 10^{-55} \times \frac{\sqrt{\pi/(2 \times 10^{14})}}{1.616 \times 10^{-35}\,\mathrm{m}} \times 3.086 \times 10^{22}\,\mathrm{m\,Mpc}^{-1} \\ &= 6.842 \times 10^{-55} \times 2.455 \times 10^{27}\,\mathrm{m}^{-1} \times 3.086 \times 10^{22}\,\mathrm{m\,Mpc}^{-1} \\ &\simeq 1.64 \times 10^{-4}\,\mathrm{Mpc}^{-1}\,. \end{aligned} \tag{A.11}$$

This confirms eq. (12) in the main text.

Step 6: Tensor-to-scalar ratio. From $H_{\mathrm{inf}} = M_{\mathrm{Pl}}\sqrt{8\pi^2/\ln D}$ (where $M_{\mathrm{Pl}} = m_{\mathrm{Pl}}/\sqrt{8\pi}$ is the reduced Planck mass):

$$P_T = \frac{2}{\pi^2}\frac{H_{\mathrm{inf}}^2}{M_{\mathrm{Pl}}^2} = \frac{2}{\pi^2} \times \frac{8\pi^2}{\ln D} = \frac{16}{\ln D}\,, \tag{A.12}$$

giving

$$r = \frac{P_T}{A_s} = \frac{16}{A_s\,\ln D}\,. \tag{A.13}$$

For $A_s = 2.1 \times 10^{-9}$ and $\ln D = 2 \times 10^{14}$: $r = 16/(2.1 \times 10^{-9} \times 2 \times 10^{14}) = 3.81 \times 10^{-5}$, confirming eq. (17).

B MCMC convergence diagnostics

We present diagnostic tests for the MCMC analysis described in section 3.

Chain statistics. The single Metropolis–Hastings chain comprises 11 681 samples with a total weight of 41 849 (mean multiplicity 3.6), corresponding to an acceptance rate of 28% and an effective sample size $N_{\mathrm{eff}} \simeq 5\,650$ (computed as $N_{\mathrm{eff}} = (\sum w_i)^2/\sum w_i^2$). The final Gelman–Rubin statistic, computed via the split-chain method, is $R - 1 = 0.012$.

Trace plots. Figure 14 shows the trace plots and running weighted means for the five key parameters: $\log_{10}(\ln D)$, α, H_0, τ, and n_s. All running means stabilise within the first $\sim 1\,000$ samples and show no systematic drift thereafter, indicating adequate mixing. The CCR parameters $\log_{10}(\ln D)$ and α exhibit broader excursions than the ΛCDM parameters, reflecting the flat posterior (the chain explores the full prior range uniformly).

Gelman–Rubin diagnostic. Figure 15 shows the split-chain $R-1$ statistic as a function of sample number. All parameters fluctuate near or below the $R-1=0.01$ target for the majority of the chain, with the final value of $R-1=0.012$ reflecting transient excursions rather than systematic non-convergence.

χ^2 trace. Figure 16 shows the χ^2 trace, confirming that the chain does not become trapped in local minima. The running mean stabilises at $\langle\chi^2\rangle = 1008.7$ with minimum $\chi^2_{\rm min} = 1002.7$.

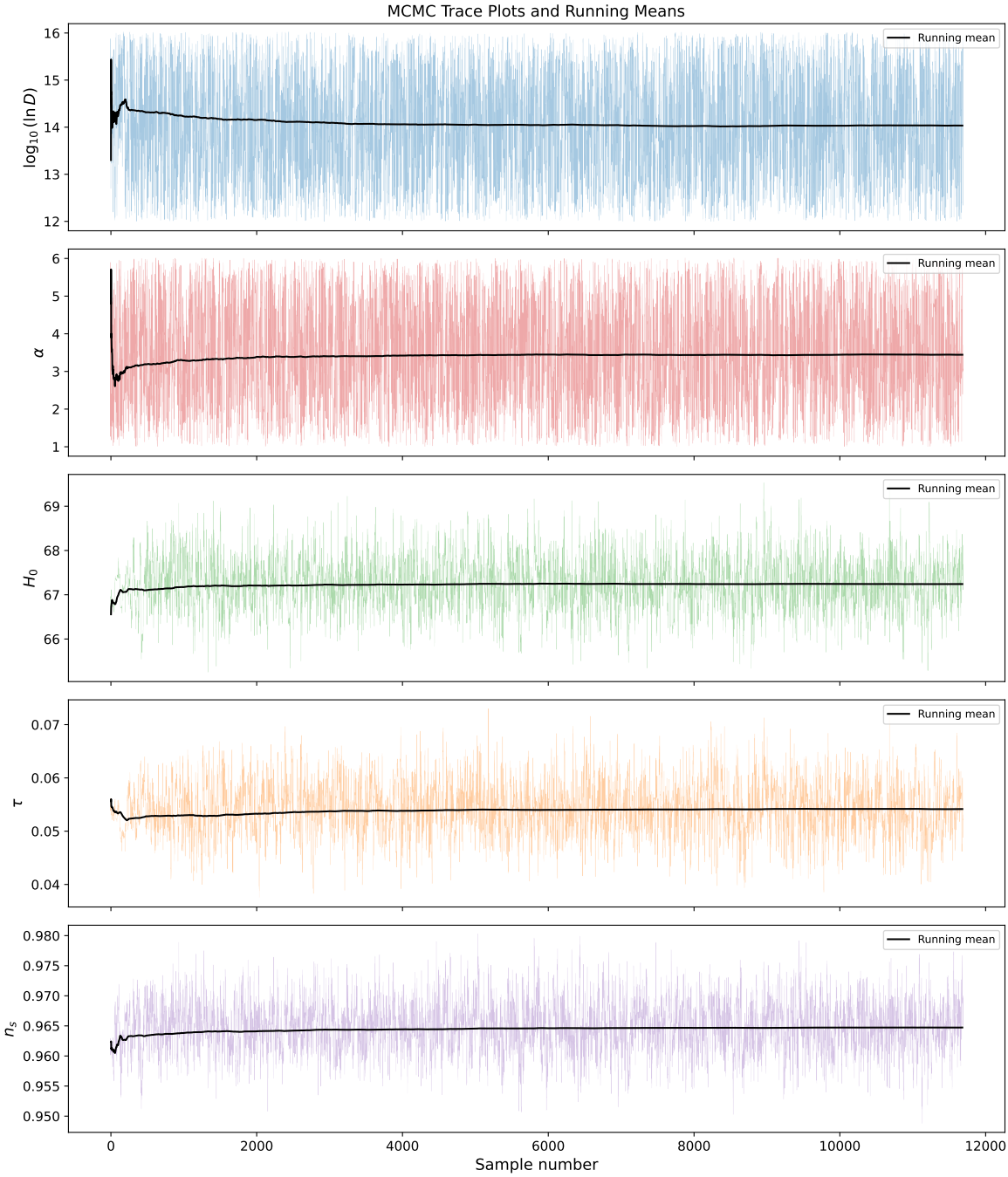

Figure 14. Trace plots and running weighted means for the five key parameters. The CCR parameters explore the full prior range uniformly, while the ΛCDM parameters converge rapidly.

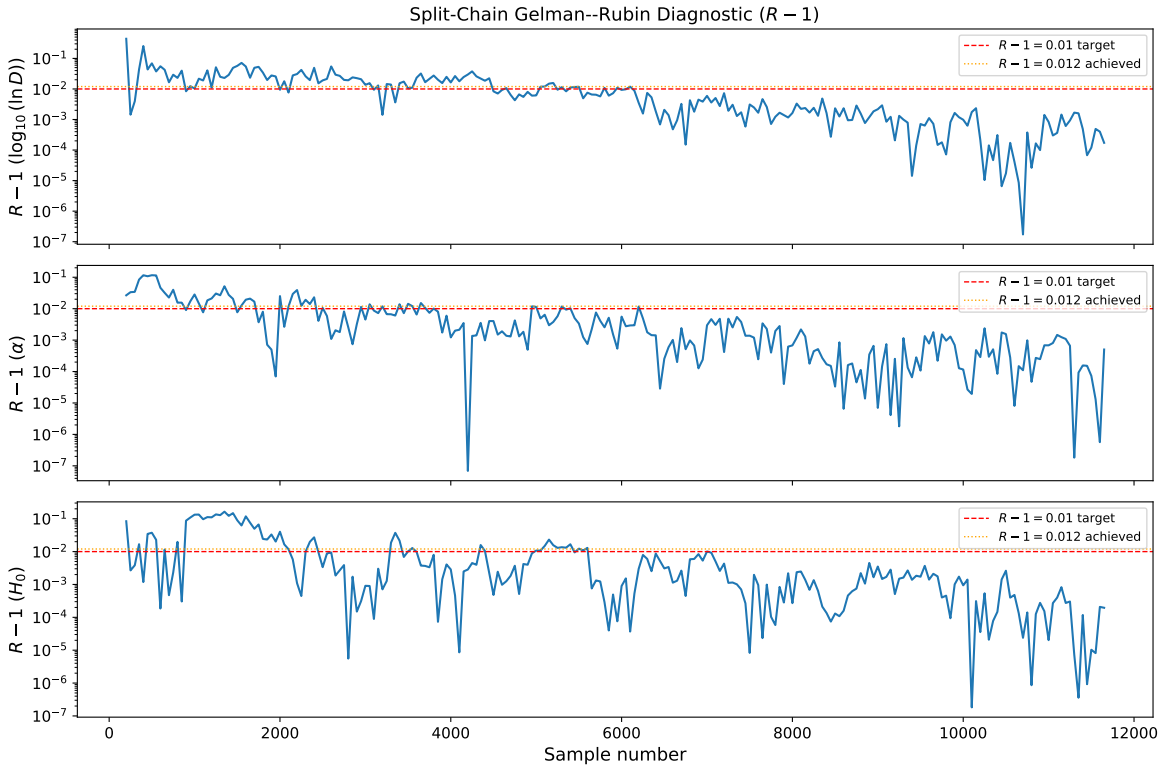

Figure 15. Split-chain Gelman–Rubin diagnostic $R-1$ as a function of sample number. The dashed line marks the $R-1=0.01$ target. All parameters fluctuate near or below $R-1=0.01$ for the majority of the chain, with a final value of $R-1=0.012$.

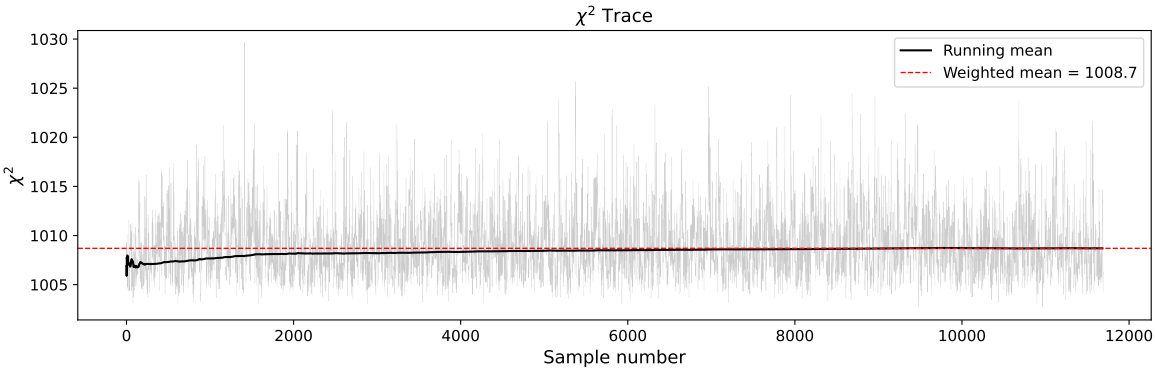

Figure 16. χ^2 trace plot. The running mean (black) stabilises at $\langle\chi^2\rangle = 1008.7$; the minimum is $\chi^2_{\rm min} = 1002.7$.

C Numerical validation

ΛCDM recovery. Figure 17 (upper panel) shows the fractional difference $|\Delta C_\ell^{TT}/C_\ell^{TT}|$ between the CCR-modified and standard ΛCDM spectra for increasing values of $\ln D$. The results confirm the expected limiting behaviour:

$\ln D$	k_c [Mpc^{-1}]	$\max\lvert\Delta C_\ell^{TT}/C_\ell^{TT}\rvert$
10^{14}	2.3×10^{-4}	2.7×10^{-1}
10^{15}	7.3×10^{-5}	1.6×10^{-2}
10^{16}	2.3×10^{-5}	2.2×10^{-4}
10^{17}	7.3×10^{-6}	1.8×10^{-6}
10^{18}	2.3×10^{-6}	2.8×10^{-11}

For $\ln D = 10^{16}$ (the upper edge of our prior range), the maximum fractional deviation is 2.2×10^{-4}, confirming that the ΛCDM limit is recovered to high precision. For $\ln D \geq 10^{17}$, the deviation drops below 10^{-5} across all multipoles.

k-grid convergence. Figure 17 (lower panel) compares the TT power spectrum computed with $n_k = 500$, 1000, and 1500 k-grid points against a reference computation with $n_k = 2000$, for $\ln D = 2 \times 10^{14}$ and $\alpha = 2$. The fractional differences are below 10^{-5} for $n_k \geq 500$, confirming that our fiducial choice of $n_k = 1500$ is more than adequate.

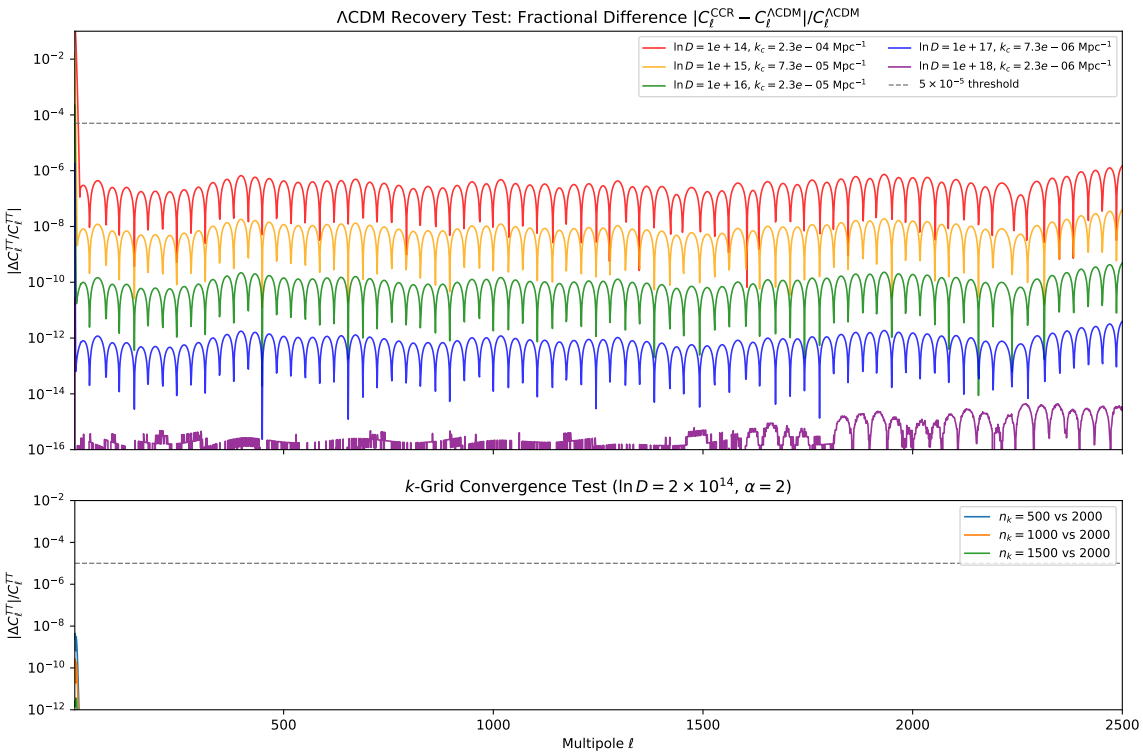

Figure 17. **Upper:** ΛCDM recovery test. Fractional difference between CCR and ΛCDM C_ℓ^{TT} for increasing $\ln D$. The dashed line marks the 5×10^{-5} threshold. **Lower:** k-grid convergence test for $\ln D = 2 \times 10^{14}$, comparing $n_k = 500$, 1000, 1500 against a reference with $n_k = 2000$. All grids agree to better than 10^{-5}.